

7 Volumes and Inequalities on Volumes of Cycles

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7.1 Curvature Inequalities

7.1.1 Bounds on Volume Elements and First Applications

7.1.1.1 The Canonical Measure and Computing it with Jacobi Fields

We start by stating, in case it is not obvious, that a Riemannian manifold M enjoys a canonical measure which can be denoted by various notations; we pick dV_M as our notation. On every tangent space we have the canonical Lebesgue measure of any Euclidean space. So the measure we are looking for is roughly the “integral” of those infinitesimal measures. If M is oriented (of dimension d) it will have a canonical *volume form*: see §§ 4.2.2 on page 166. The measure is the “absolute value” of that volume form. In any coordinates $\{x_i\}$ with the Riemannian metric represented by g_{ij} , the canonical measure is written

$$dV_M = \sqrt{\det(g_{ij})} dx_1 \dots dx_d$$

where $dx_1 \dots dx_d$ is the standard Lebesgue measure of $\mathbb{E}^d$. Similarly, the volume form of an oriented Riemannian manifold is

$$\omega = \sqrt{\det(g_{ij})} dx_1 \wedge \cdots \wedge dx_d .$$

Under scaling $g \rightarrow \lambda g$ the canonical measure undergoes

$$dV_{(M, \lambda g)} = \lambda^{d/2} dV_{(M, g)} .$$

For a domain $D \subset M$ we will use the notation $\text{Vol}(D)$ for its measure. When M is oriented, $\text{Vol}(M) = \int_M \omega$. Any submanifold of a Riemannian manifold inherits a Riemannian metric and from it a canonical measure. In particular, a compact submanifold will have a volume. In the sequel we will use the word *volume* whatever the dimension to avoid verbiage; but it should be clear that for curves volume is *length* and for surfaces it is *area*.

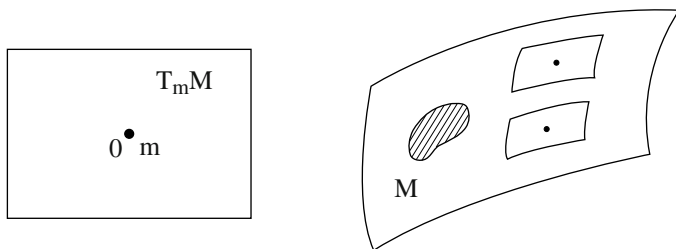


Fig. 7.1. The measure is the “integral” of the infinitesimal measures

Most often volumes in Riemannian manifolds are computed by the following scheme. One fills up a domain of the d dimensional manifold M by a $(d - 1)$ parameter family geodesics, so that the d^{th} dimension is nothing but the arc length s along the geodesics. By the Fubini theorem the volume of the domain under consideration will be the integral with respect to s of the integral with respect to the $d - 1$ parameters of the infinitesimal $(d - 1)$ dimensional volume element of the “tube”. This is what we have to compute. It is reasonable to start at time $s = 0$ with initial point n running through some hypersurface H (our $d - 1$ parameters) and the corresponding geodesics $\gamma = \gamma_n$ normal to H at n .

Since an infinitesimal variation of geodesics is a Jacobi field Y (see equation 6.11 on page 248), to answer our question we need to know the initial conditions $Y(0)$ and $Y'(0)$ for such Jacobi field. The value $Y(0)$ is nothing but some unit vector $e \in T_n H$ and by the same trick as in equation 6.8 on page 243 the value of $Y'(0)$ will be equal to the parallel transport derivative of the vector field along the curve $c_e(\alpha)$ of H (with $c'_e(0) = e$) given by the initial speed vector $c'_e(0)$ of the family of geodesics normal to H at $c_e(\alpha)$.

This is for a one parameter infinitesimal displacement of γ . Now we need $d - 1$ of them. Pick up a orthonormal basis $\{e_i\}_{i=1, \dots, d-1}$ of $T_n H$. To each e_i is associated a Jacobi field Y_i along γ with initial conditions $Y_i(0) = e_i$ and

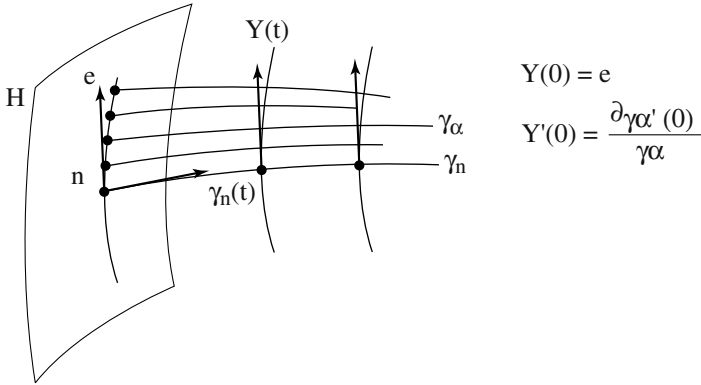


Fig. 7.2. Measuring volume by extending normal geodesics off of a hypersurface

$$Y'_i(0) = h_i = \left. \frac{\partial \gamma_\alpha}{\partial \alpha'} \right|_{\alpha=0}$$

where (as was said above) the derivative has to be taken using the notion of parallel transport (see § 15.4 on page 701) and where γ_α denotes the geodesic starting from $c_e(\alpha) \in H$ and proceeding orthogonally to H . Then the picture in figure 7.3 on the next page makes intuitive that we have:

Lemma 101

$$dV_M(\gamma(t)) = \det(Y_1(t), \dots, Y_{d-1}(t)) dt \wedge dV_H(n)$$

where the determinant is the canonical one associated to the $(d-1)$ dimensional Euclidean subspace of $T_{\gamma(t)}M$ orthogonal to $\gamma'(t)$.

This is in fact nothing but the change of variables formula.

Now suppose we want to compute the volume of a ball centered around a point $m \in M$. This time the natural $(d-1)$ parameter family of geodesics is that of all geodesics emanating from the center m , and its natural parameterization is the unit sphere $U_m M$. This time we have no hypersurface H , but we replace it by $U_m M$. Let γ be a geodesic starting from m with speed vector $\gamma'(0) = e \in U_m M$ and complete e into an orthonormal basis

$$\text{span}\{e_i\} \quad i = 1, \dots, d-1$$

of the tangent space $T_m M$. Denote by Y_i the Jacobi field along γ with initial conditions

$$Y_i(0) = 0 \quad \text{and} \quad Y'_i(0) = e_i.$$

Then the analogue of lemma 101 is:

$$dV_M(\gamma(t)) = \det(Y_1(t), \dots, Y_{d-1}(t)) dt d\sigma \quad (7.1)$$

where $d\sigma$ is the canonical measure on the sphere $U_m M$.

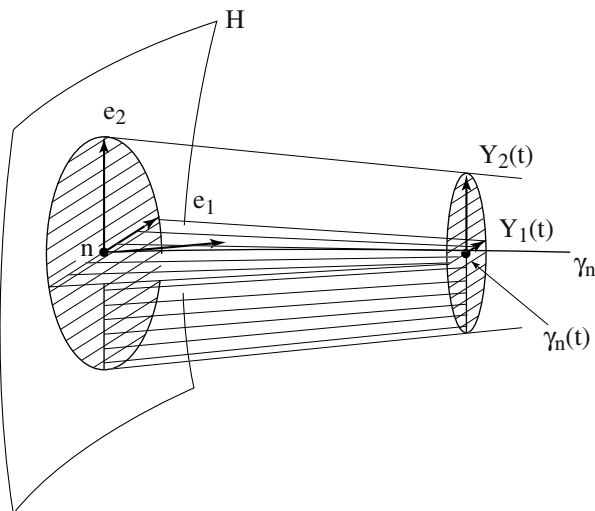


Fig. 7.3. Calculating the volume form

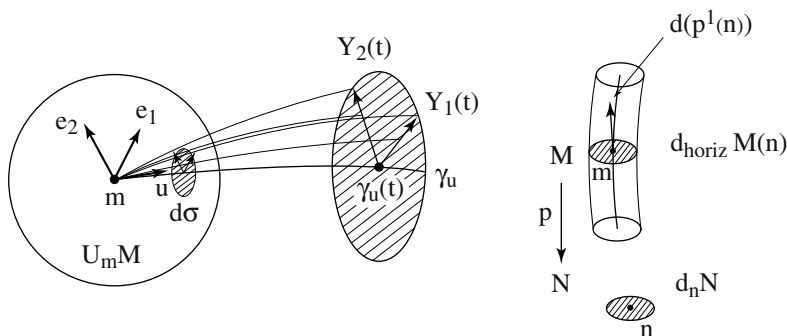


Fig. 7.4. Computing volumes of small balls

Note 7.1.1.1 There is at least one interesting type of volume computation not covered by the above technique, that of Riemannian submersions (see §§4.3.6). We leave as an exercise to prove that for a Riemannian submersion

$$p : (M, g) \rightarrow (N, h)$$

the volume elements are:

$$dV_M(m) = dV_{p^{-1}(n)}(m) dHm \quad (7.2)$$

where H is the hyperplane in $T_m M$ perpendicular to the tangent space of the fiber $p^{-1}(n)$.

In particular,

$$\text{Vol}(M) = \int_N \text{Vol } p^{-1}(n) dV_N(n)$$

In many cases, in particular when the fibers are all totally geodesic submanifolds, their volume is constant. Then

$$\text{Vol}(M) = \text{Vol}(\text{fiber}) \cdot \text{Vol}(N) .$$

This is the case for example for the submersions from spheres to the $\mathbb{K}\mathbb{P}^n$ seen in §§4.1.3. This is one way to compute the volume of the $\mathbb{K}\mathbb{P}^n$ (except for $\mathbb{C}\mathbb{P}^2$) if one knows the volumes of spheres; we will find these next. $\blacklozenge$

7.1.1.2 Volumes of Standard Spaces

Formula 7.1 on page 301 enables us to compute the volume of balls of any radius in Euclidean spaces, spheres of various radii, hyperbolic spaces of any constant curvature and in the $\mathbb{K}\mathbb{P}^n$. It is enough to remember formula 6.14 on page 251 to find volumes of Euclidean, spherical and hyperbolic balls. For balls in the $\mathbb{K}\mathbb{P}^n$, one will need formula 4.36 on page 209 and need to pick an adapted basis. In particular, one can compute the (total) volume of those of the preceding spaces which are compact: spheres and $\mathbb{K}\mathbb{P}^n$. Details are given, partly or completely, in III.H of Gallot, Hulin & Lafontaine 1990 [542], 3.3 of Chavel 1993 [326], Berger 1965 [151] and Sakai 1996 [1085]. With the notation

$$\sigma_d = \text{Vol } S^d$$

the volumes of balls are

$$\text{Vol } B(S^d, r) = \sigma_{d-1} \int_0^r \sin^{d-1}(t) dt$$

where since the sphere (as well all the spaces below) is homogeneous, we need not make precise which point is the center of the ball.

In particular, σ_d is obtained by induction using the value of the classical integral on the right. We then get:

$$\begin{aligned} \text{Vol } S^{2n} &= \frac{2^{n+1} \pi^n}{1 \cdot 3 \dots (2n-1)} \\ \text{Vol } S^{2n+1} &= \frac{2\pi^{n+1}}{n!} \end{aligned}$$

or in one shot, with the classical Γ function:

$$\text{Vol } S^d = \frac{2\pi^{(d+1)/2}}{\Gamma(\frac{d+1}{2})} .$$

In hyperbolic space,

$$\text{Vol } B\left(\text{Hyp}^d(-k^2), r\right) = \sigma_{d-1} \int_0^r \left(\frac{\sinh(kt)}{k}\right)^{d-1} dt$$

And for the $\mathbb{KP}^n$ of diameter $\pi/2$:

$$\text{Vol } B(\mathbb{CP}^n, r) = \sigma_{2n-1} \int_0^r \frac{\sin(2t)}{2} \sin^{2n-2}(t) dt$$

and in particular

$$\begin{aligned} \text{Vol } \mathbb{CP}^n &= \frac{\pi^n}{n!} \\ &= \frac{\sigma_{2n+1}}{\sigma_1}. \end{aligned}$$

$$\text{Vol } B(\mathbb{HP}^n, r) = \sigma_{4n-1} \int_0^r \left(\frac{\sin(2t)}{2}\right)^3 \sin^{4n-4}(t) dt$$

and in particular

$$\begin{aligned} \text{Vol } \mathbb{HP}^n &= \frac{\pi^{2n}}{(2n+1)!} \\ &= \frac{\sigma_{4n+3}}{\sigma_3} \end{aligned}$$

$$\text{Vol } B(\mathbb{CaP}^2, r) = \sigma_{15} \int_0^r \left(\frac{\sin(2t)}{2}\right)^7 \sin(8t) dt$$

and in particular

$$\text{Vol } \mathbb{CaP}^2 = \frac{6\pi^8}{11!}$$

It is amusing to observe that this volume is equal to σ_{23}/σ_7 although there is no fibration $S^{23} \rightarrow \mathbb{CaP}^2$ as mentioned on page 155.

7.1.1.3 The Isoperimetric Inequality for Spheres

In the 1940's Schmidt proved the isoperimetric inequality for spheres of any dimension (endowed with the constant curvature metric) as well as for hyperbolic spaces. Great simplification comes from the fact that those spaces have as many as possible hyperplane symmetries¹ (see theorem 40 on page 208). The result is what one expects:

¹ An exercise for the reader: prove that this multitude of isometric involutions characterizes these spaces.

Theorem 102 (Spherical isoperimetric inequality) *Among all domains of a given volume living inside a sphere, the metric balls are precisely the domains whose boundary has the minimum volume.*

See an application to the concentration phenomenon in note 7.1.1.2. A detailed proof is given (for example) in 6.3 of Chavel 1993 [326]; see also IV.H of Gallot, Hulin & Lafontaine 1990 [542] and chapter VI of Sakai 1996 [1085]; but the basic reference today is Burago & Zalgaller 1988 [283].

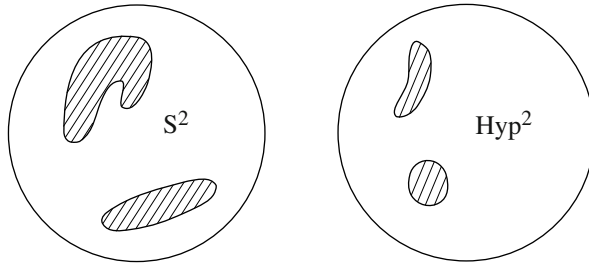


Fig. 7.5. The isoperimetric inequality on the sphere and in hyperbolic space

Note 7.1.1.2 (On the volumes of balls in spheres) Looking at the graph of the function $\sin^{d-1}(t)$ for large d we see that the volume of a ball in S^d grows extremely slowly with the growth radius for small radii, and then changes drastically around the equator. This is called the *concentration phenomenon*: the sphere is concentrated around its equator in the sense that a very small tubular neighborhood of an equator has almost full measure. From this and from the spherical isoperimetric inequality (theorem 102) one can deduce this surprising phenomenon: on a sphere of very large dimension, any function is very close to its mean value on a subset of very large measure. To see this, one employs lemma 101 on page 301. This is basic in proving Dvoretzky's theorem for symmetric convex bodies in Euclidean spaces: for every ε a convex body admits ε -almost spherical sections of a suitable dimension (of logarithmic growth in d). References for this fascinating subject: Lindenstrauss 1992 [869], Milman 1992 [920], Pisier 1989 [1030], Lindenstrauss & Milman 1993 [870], Berger 2003 [173]. For a general setting for this story of "geometric probabilities", see Gromov's *mm* spaces in §14.6.

The values given above for the σ_d show also that σ_d goes to zero very quickly (use Stirling's formula to see that σ_d behaves like $d^{-d/2}$). An amusing exercise is to compute the first d for which $\sigma_d < 1$. ♦

7.1.1.4 Sectional Curvature Upper Bounds

We simply mix Rauch's inequality (proposition 74 on page 259) together with equation 7.1 on page 301 above to get a simple control result:

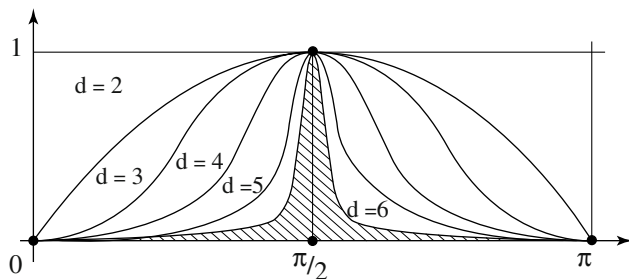


Fig. 7.6. The graph of $\sin^{d-1}(t)$ for various d

Theorem 103 Assume $K \leq \Delta$ everywhere on a complete d dimensional Riemannian manifold M and that $\exp_m$ is a diffeomorphism on $B(m, r)$. Then

$$\text{Vol } B(m, r) \geq \text{Vol } B(S^d(\Delta), r)$$

To get a complete proof one only needs the following trick: at a given t in equation 7.1 on page 301 replace $\{Y_i(t)\}$ by an *orthogonal* basis $\{Z_i\}$ obtained by taking an orthogonal transformation of the orthogonal complement of the velocity $\gamma'(0)$ in $T_m M$. Then the determinant

$$\det(Y_1(t), \dots, Y_{d-1}(t))$$

is equal to the product

$$\det(Z_1(t), \dots, Z_{d-1}(t)) = \prod_{i=1}^{d-1} \|Z_i\|$$

and

$$\|Z_i\| \geq \frac{\sinh(\sqrt{\Delta}t)}{\sqrt{\Delta}}$$

by proposition 74. An analogous trick will be used in the proof of lemma 106 on page 309.

The inequality in theorem 103 was used in 1968 by Milnor (see also Švarc 1955 [1170] and Karcher 1989 [780]) to get:

Theorem 104 (Milnor 1968 [924]) *The fundamental group of compact manifold of negative curvature has exponential growth.*²

Definition 105 *The growth of a finitely generated group with fixed choice of generators is the function $\gamma(s)$ which tells us the number of elements of the group which can be expressed as words in those generators with length smaller than s . The growth is said to be polynomial if one has*

² It is easy to see that the fundamental group of a compact manifold is finitely generated.

$$\gamma(s) \leq s^k$$

for some integer k , and all sufficiently large s , while the growth is exponential if one has

$$\gamma(s) \geq a^s$$

for some number $a > 1$. Both of these notions make sense because they are invariant under changes of the choice of generators.

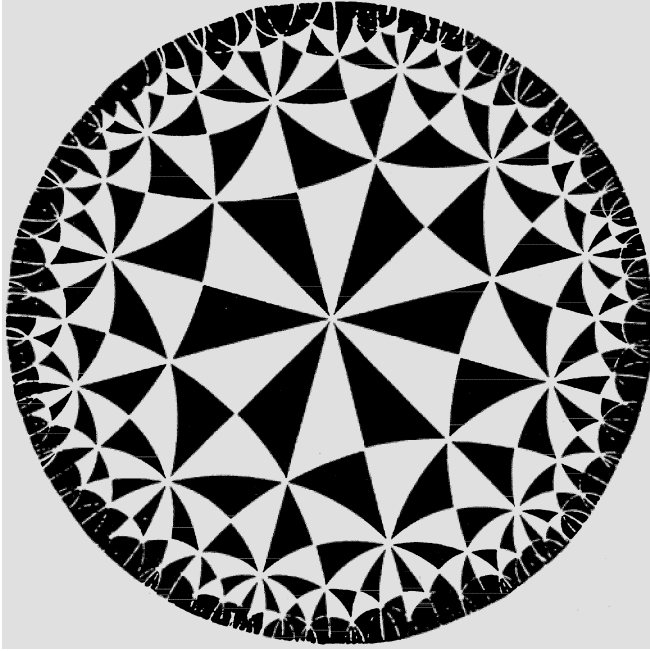


Fig. 7.7. A hyperbolic tiling

We sketch the proof, which is very geometric. We consider the universal Riemannian covering $\tilde{M} \rightarrow M$ (which is not compact, by the von Mangoldt–Hadamard–Cartan theorem 72 on page 255). Being simply connected and of negative curvature we know from that theorem that we can apply theorem 103 on the preceding page to it for any real number r and get that for any point $\tilde{m} \in \tilde{M}$:

$$\text{Vol } B(\tilde{m}, r) > a \exp(br)$$

for some constants $a, b > 0$.

Fix $\tilde{m}$ in $\tilde{M}$ and set

$$N = B(\tilde{m}, \text{diam } M) .$$

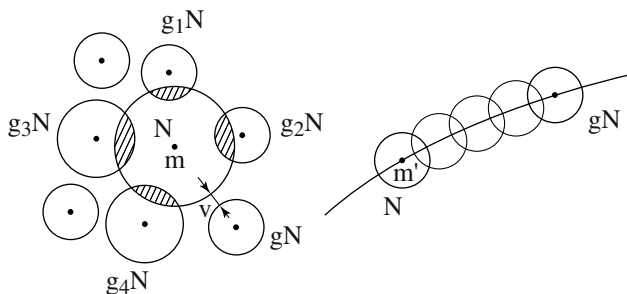


Fig. 7.8. Proof of theorem 104 on page 306

Recall (see §§4.3.3) that the elements of the fundamental group $\pi_1(M)$ act on $\tilde{M}$ by isometries. Introduce the subset of $\pi_1(M)$ made up of the g such that gN intersects N . These g build up a finite subset F of $\pi_1(M)$. The other gN are always at a positive minimum distance ν from N . If the distance from $\tilde{m}$ to some gN is such that

$$d(\tilde{m}, gN) < n\nu + \text{diam } M$$

for some integer n then the picture below will show that g can be written as a product

$$g = g_1 \cdots g_n$$

of elements of F . This proves first that $\pi_1(M)$ is generated by F . Second, it proves that the set of the gN build up by the g of word length not larger than n will certainly cover the ball $B(\tilde{m}, n\nu + \text{diam } M)$ so that:

$$\gamma(s) \text{Vol } N \geq \text{Vol } B(\tilde{m}, n\nu + \text{diam } M)$$

and we are done via equation 7.1.1.4 on the previous page.

7.1.1.5 Ricci Curvature Lower Bounds

Without any trick, since we always have

$$\det(Y_i) \leq \Pi_i \|Y_i\|,$$

one would clearly bet that a lower bound $K > \delta$ would provide an upper bound on the volume of balls, through theorem 62 on page 243. But in 1963 Bishop made a striking discovery to the effect that a Ricci curvature lower bound is enough to get an optimal upper bound on the volume of balls. According to Gromov such a result is outlined already in Lévy 1951 [861] but it seems to have been unnoticed thereafter.

We will first state a more general inequality due to Heintze and Karcher in 1978 and then give two consequences of it (for reference see 7.5 of Chavel 1993 [326], and also 4.21 in Gallot, Hulin & Lafontaine 1990 [542]):

Lemma 106 (Heintze & Karcher) *As in lemma 101 on page 301 let*

$$\{Y_i(t)\}_{i=1,\dots,d-1}$$

be Jacobi fields along some geodesic γ and set

$$F = \det(Y_1 \dots, Y_{d-1})$$

and

$$f = F^{1/d-1}.$$

Then f satisfies:

$$f'' + \frac{\text{Ricci}(\gamma')}{d-1} f \geq 0$$

under the proviso that f does not vanish (except possibly at the origin).

There is a straightforward computation plus a trick analogous to that of theorem 103 on page 306. Computation yields:

$$\begin{aligned} f'' &= \frac{2-d}{(d-1)^2} f F^{-2} \left(\sum_i \det(Y'_i) \right)^2 \\ &\quad + \frac{1}{d-1} f F^{-1} \left(\sum_i \det(Y'_i, Y'_j) + \sum_i \det(Y''_i) \right) \end{aligned}$$

with the obvious notations for the determinants; for example Y'_i should appear at the i^{th} place and at all the other places one puts Y_i etc. The last term is easy and classically by linear algebra yields the value

$$\frac{1}{d-1} f \text{Ricci}(\gamma').$$

But the two other terms are a mess in the general expression for the $Y_i(t)$. To clear up things one must first fix some time t . By an orthogonal transformation of $T_{\gamma(t)}H$ one can change the sets

$$\{Y_i(t)\}, \{Y'_i(t)\}$$

into sets

$$\{A_i\}, \{B_i\}$$

such that

$$B_i = \lambda_i A_i.$$

This is possible because the bilinear form given by the scalar products

$$\langle Y'_i(t), Y_j(t) \rangle$$

is symmetric. This is not obvious; it comes from the fact that the Jacobi field equation 6.11 on page 248 and the symmetries of the curvature tensor (see equation 4.28 on page 203) implies that

$$(\langle Y', Z \rangle - \langle Y, Z' \rangle)' = 0$$

so that we need only to insure symmetry at time $t = 0$ which can be done for example taking for the e_i the eigendirections of the second fundamental form.

Note that the determinant is not changed so that we are left with two terms:

$$\left(\sum_i \lambda_i \right)^2 \quad \text{and} \quad \sum_{i,j} \lambda_i \lambda_j .$$

The total result turns out to be

$$-\frac{1}{(d-1)^2} \sum_{i,j} (\lambda_i - \lambda_j)^2$$

proving the theorem.

Note 7.1.1.3 Beware that the above trick is only valid when the full action of $SO(d-1)$ is available. This is the case for the set of geodesics emanating from a given point or that of geodesics emanating orthogonal to a given hypersurface. It does not work, for example, for geodesics emanating orthogonally from a curve (as in theorem 90 on page 273 for example). In such a case one needs more assumptions that only controlling the Ricci curvature. ♦

The basic consequence of theorem 106 on the preceding page is:

Theorem 107 (Bishop–Gromov 1963,1999 [196, 633]) *Assume that*

$$\text{Ricci} \geq (d-1)\delta$$

where δ is any real number. Starting from any point $m \in M$ the function

$$r \rightarrow \frac{\text{Vol } B(m, r)}{\text{Vol } B(\mathbb{S}^d(\delta), r)}$$

is nowhere increasing. In particular for any $r \geq 0$ we have

$$\text{Vol } B(m, r) \leq \text{Vol } B(\mathbb{S}^d(\delta), r) .$$

We have only to integrate in polar coordinates centered at m and apply equation 7.1 on page 301: the determinant will be bounded by f^{d-1} where f satisfies

$$f'' + \delta f > 0 .$$

Looking at the limit when $r \rightarrow 0$, the initial condition yields then

$$f \leq f_\delta$$

where f_δ is the corresponding function in the space form $\mathbb{S}^d(\delta)$.

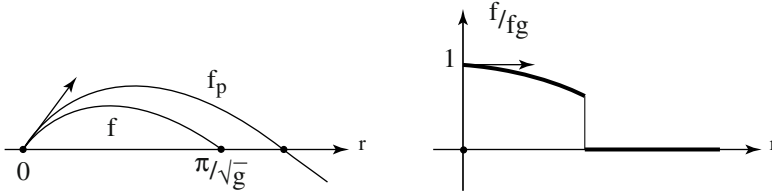


Fig. 7.9. Continue with 0 after the cut value

This was Bishop's result. Now one has to wonder about the proviso in lemma 106 on page 309 which concerns conjugate points. But it is irrelevant here because we are looking for lower bound. On every geodesic starting from m one has only to go no longer than the cut-value (see §§6.5.1) and one knows that before the cut value the function f will never vanish from theorem 86 on page 268. After the cut value there is only zero to integrate. This was the contribution of Gromov, which will turn out to be essential in the future, first for getting metric bounds from lower Ricci curvature without wondering about injectivity radius, and second for getting lower bounds for the volume of balls when the total volume

$$\text{Vol } M = \text{Vol } B(m, \text{diam } M)$$

(for any point m) is bounded (this is because of the non-increasing property in theorem 107 on the facing page see equation 12.2 on page 626).

In fact, if this is not obvious at first glance, this is Gromov's philosophy as explained in 5.31 and in the Appendix B (see B.2) of Gromov 1999 [633]; we will see in the references quoted just above that what is at the root of the proofs is the:

Theorem 108 (The doubling property) *In a Riemannian manifold of dimension d with nonnegative Ricci curvature, any ball of radius $2R$ can be covered by a collection of 4^d balls of radius R . If we only know that $\text{Ricci} \geq (d-1)\delta$, then there is an analogous inequality, but only valid for "small balls" and with some constant depending only on d and δ .*

We have only to use the covering trick (lemma 125 on page 333): pack in the given ball of radius $2R$ as many balls of radius $R/2$ as possible. Then the balls of same centers and radius R must cover the initial ball. But now use the Bishop–Gromov theorem 107 on the preceding page and remember that the volume of a Euclidean ball of radius ρ is proportional to ρ^d . This shows that there cannot be more than 4^d disjoint balls of radius $R/2$ within a ball of radius $2R$. For the doubling property in a completely different context, see Toro 1997 [1197].

The natural analogue of theorem 104 on page 306 is:

Theorem 109 (Milnor 1968 [924]) *Let M be a complete Riemannian manifold with non-negative Ricci curvature (not necessarily compact). Then the growth function of any finitely generated subgroup of the fundamental group $\pi_1(M)$ satisfies*

$$\gamma(s) \leq as^d$$

for some positive constant a .

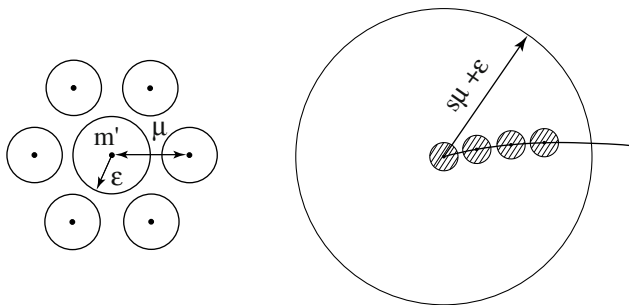


Fig. 7.10. Ball packing and growth of the fundamental group

Let us look once again at the universal Riemannian covering $\tilde{M}$ which has nonnegative Ricci curvature because M does. Pick some point $\tilde{m}$ in $\tilde{M}$. Recall that the fundamental group $\pi_1(M)$ acts as isometries on $\tilde{M}$. There is $\varepsilon > 0$ such that $B(\tilde{m}, \varepsilon)$ does not meet any other $B(g\tilde{m}, \varepsilon)$ for g running through $\pi_1(M)$ (except the identity). Let us also introduce the supremum μ of the distances $d(\tilde{m}, g_i\tilde{m}')$ when g_i runs through the generators of a finitely generated subgroup. Then the ball $B(\tilde{m}, s\mu + \varepsilon)$ will contain at least $\gamma(s)$ disjoint balls of radius ε to the effect that

$$\gamma(s) \text{Vol } B(\tilde{m}, \varepsilon) \leq \text{Vol } B(\tilde{m}, s\mu + \varepsilon)$$

and theorem 108 on the previous page (the doubling property) implies the desired assertion.

Note 7.1.1.4 (About Milnor's result) A deep theorem of Gromov says that a group of polynomial growth is a discrete subgroup of a nilpotent Lie group: see Gromov 1981 [614] or Gromov 1993 [629]. We will come back to this in theorem 352 on page 598.

Milnor conjectures in his article that the fundamental group is automatically finitely generated (for complete noncompact manifolds of course): to our knowledge it is still an open question.

Note that the doubling property theorem 108 is a refinement of Myers' theorem 63 on page 245. Also note that one cannot do better: take flat tori for example. Compare with Bochner's theorem 345 on page 594. ♦

Note 7.1.1.5 A priori one cannot expect a lower bound for the volumes of balls with only an upper bound on the Ricci curvature: in fact there are manifolds with identically zero Ricci curvature (see Besse 1987 [183]) and for which there is no lower bound in comparison with Euclidean space for the volume of balls. There is one exception (besides of course dimension two): this is dimension three. The reason is that in dimension three Ricci curvature and sectional curvature are equivalent (an exercise left to the reader): see Eschenburg & O'Sullivan 1980 [496]. This was *stricto sensu*. But as remarked above, if for example we know the total volume, then we can hope to control volumes of balls. ♦

We come now to the second application of lemma 106 on page 309. To express it we need to first extend to general Riemannian manifolds the notion of *second fundamental form* of a hypersurface. As in §§1.6.3 this is a quadratic form on the tangent space $T_n H$ at n of a hypersurface H of our Riemannian manifold M . It tells us how much H differs from the hypersurface generated by the various geodesics starting from n and tangent to H . With the notation of §§7.1.1.1 it is nothing but the quadratic form given by the scalar products

$$\langle e_i, h_i \rangle .$$

The *mean curvature* of H at n is by definition equal to the trace of the second fundamental form of H at n divided by $d - 1$ and is denoted by η .

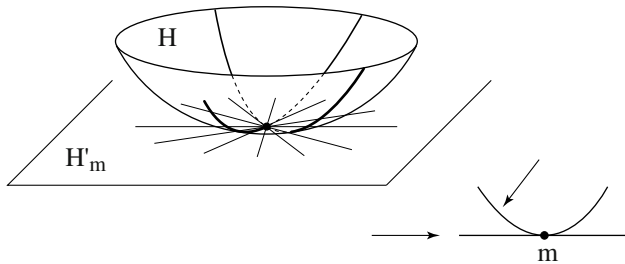


Fig. 7.11. Comparing a hypersurface with the hypersurface formed from ambient space geodesics emanating out of a point and tangent to our original hypersurface

The mean curvature of hypersurface in general Riemannian manifolds has the same property that it has in Euclidean spaces: formula 9 on page 59 extends without modification.³

Lemma 110 (First variation equation) *For a normal variation of length f of a piece H of hypersurface, the first derivative of the volume is equal to*

³ Just recall that the mean curvature is only determined up to a sign but in theorem 111 on the next page below a preferred normal will be given by $\gamma'(0)$.

$$\int_H f \eta dV_H .$$

In particular, assume that D is a compact domain in M with boundary ∂D and that ∂D has minimal volume among all domains close to D having the same volume as D . Then ∂D has constant mean curvature.

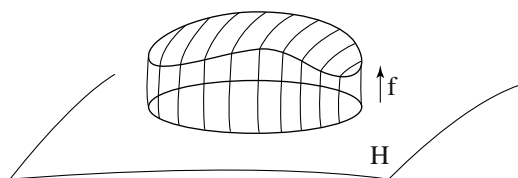


Fig. 7.12. The first variation of a hypersurface

We now come to the evaluation of the volume equation in lemma 101 on page 301:

Theorem 111 (Heintze–Karcher, 1978) *Assume $\text{Ricci} > (d-1)\delta$ and consider a pencil of geodesics normal to a hypersurface H neighboring some geodesic γ . Provided that the determinant in lemma 101 on page 301 does not vanish from 0 to t then*

$$dV_M(\gamma(t)) < f_\delta^{d-1}(t)$$

where η is the mean curvature of H at $\gamma(0)$ and f_δ is the standard function

$$f_\delta = \begin{cases} \cos(t\sqrt{\delta}) + \frac{\eta}{\sqrt{\delta}} \sin(t\sqrt{\delta}) & \text{when } \delta > 0 \\ 1 + \frac{\eta}{t} & \text{when } \delta = 0 \\ \cosh(t\sqrt{-\delta}) + \frac{\eta}{\sqrt{-\delta}} \sinh(t\sqrt{-\delta}) & \text{when } \delta < 0. \end{cases}$$

For the proof we first remark that f and f_δ have the same initial conditions

$$f(0) = f_\delta(0) = 1$$

and

$$f'(0) = f'_\delta(0) = \eta$$

by the very construction of the Y_i and the definition of the mean curvature. Then we proceed as in on page 121 by the Sturm-Liouville technique. As an exercise, compare the above with the proof in §§1.6.8.

7.1.2 Bounding the Isoperimetric Profile with the Diameter and Ricci Curvature

7.1.2.1 Definition and Examples

The isoperimetric inequality in $\mathbb{E}^d$ is a fundamental geometric notion and also has many applications. We gave one in detail: the Faber–Krahn inequality 1.22 on page 81. We saw also in §§§7.1.1.3 that there is an isoperimetric inequality in the standard space forms, the spheres and hyperbolic spaces, so that it is natural to ask

Question 112 *Is there an isoperimetric inequality in a general Riemannian manifold?*

Let us say first that the state of affairs today is far from what we hope for. For example one would expect the domains optimal for isoperimetry in the symmetric spaces of rank one would be the metric balls. This is extremely reasonable since those spaces are precisely the isotropic (or two-point homogeneous) ones (see §§6.6.2). We will see below that is false for compact spaces, but that there is a very reasonable conjecture. But for the noncompact, simply connected ones, one would expect the balls to be optimal. However this is a completely open problem today, starting with the $\text{Hyp}_{\mathbb{C}}^n$.

To be more precise, we consider domains D in a Riemannian manifold M with a smooth boundary ∂D . As in the case of the sphere we cannot in general hope to have a universal bound for the dimensionless ratio

$$\frac{\text{Vol}(\partial D)^d}{\text{Vol}(D)^{d-1}} .$$

This was possible for Euclidean spaces essentially because they admit nonisometric similarities. We need a concept where $\text{Vol}(D)$ enters explicitly and is normalized in some way.

Definition 113 *Take M a compact Riemannian manifold. For every real number*

$$\beta \in [0, 1]$$

we define $h(\beta)$ as

$$h(\beta) = \inf_D \left\{ \frac{\text{Vol}(\partial D)}{\text{Vol}(M)} \mid \text{Vol}(D) = \beta \text{Vol}(M) \right\} .$$

The function $\beta \mapsto h(\beta)$ is called the isoperimetric profile of the Riemannian manifold M . The more crude Cheeger’s constant h_c was introduced in Cheeger 1970 [329] and is defined as above, but in terms of domains D with

$$\text{Vol}(D) \leq \frac{1}{2} \text{Vol}(M) .$$

It is then equal to

$$h_c = \inf_{\beta \leq \frac{1}{2}} h(\beta) .$$

This profile can be quite wild, for example for the surface in figure 7.13.

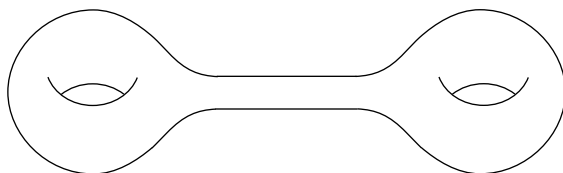


Fig. 7.13. Study $h(\beta)$ for this surface

The isoperimetric inequality 102 on page 305 for spheres and the formula in §§7.1.1.2 yields the isoperimetric profile for sphere. We suggest that you perform some computation and drawings, starting with two dimensions.

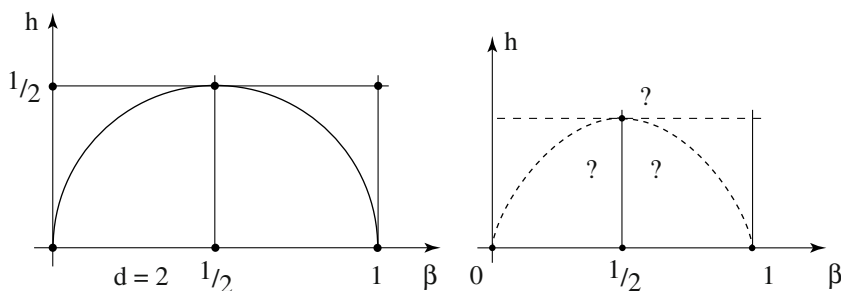


Fig. 7.14. Compute and draw isoperimetric profiles for dimensions 3, 4, ...

Let us now try to compute the isoperimetric profile of the real projective plane $\mathbb{RP}^2$, equipping it with the standard metric of constant curvature (see §§4.3.3.2). We follow Gromov's technique as in §§1.6.8. For a given β , we look for a domain D of area equal to

$$\beta \operatorname{Vol} \mathbb{RP}^2 = 2\pi\beta$$

and with smallest possible length for ∂D . We apply lemma 110 on page 313: the curve ∂D has constant curvature. In $\mathbb{RP}^2$ this implies that ∂D is a geodesic or a circle, i.e. the projection of a small circle of the sphere S^2 . To see this, show that in S^2 the curves of constant curvature are small circles and use the fact that

$$S^2 \rightarrow \mathbb{RP}^2$$

is a local isometry.

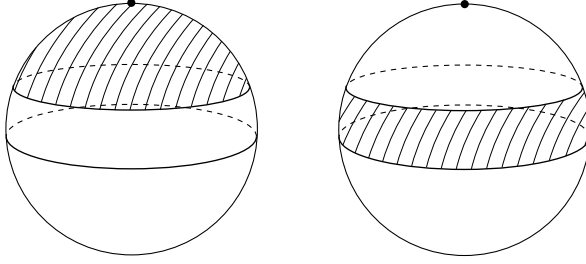


Fig. 7.15. Working in $\mathbb{RP}^2$

As expected, at least for small values of β , the optimal domains are those D whose boundaries have constant curvature, which could be a disk, or could very well be the tubular neighborhood of a geodesic. The picture in figure 7.15 represents the story on $\mathbb{RP}^2$ lifted up to the sphere. Down in $\mathbb{RP}^2$ such a domain is a Möbius band. But its complement is an ordinary disk and their boundaries have the same length. So finally the isoperimetric profile of $\mathbb{RP}^2$ is as indicated in figure 7.16.

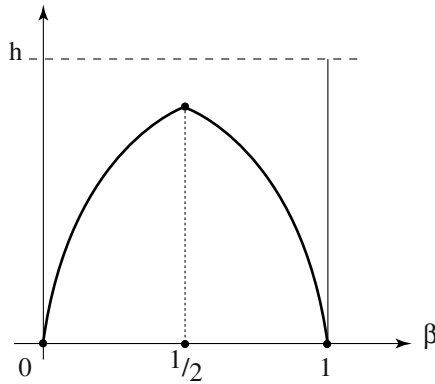


Fig. 7.16. The isoperimetric profile of the real projective plane

We now look at the square flat torus of two dimensions. Here again candidates for boundaries are ordinary plane circles or straight lines. But notice that when we look at a large disk, of radius $r > 1/\pi$, the band of width δ has a better isoperimetric ratio: the band has isoperimetric ratio $2/\delta$ while the disk has $2\pi r/\pi r^2$. Since disks and strips are the only ones with constant curvature boundary, the profile for the square T^2 is as pictured in figure 7.17 on the next page.

As an exercise, work out the case of a square Klein bottle and some other flat tori.

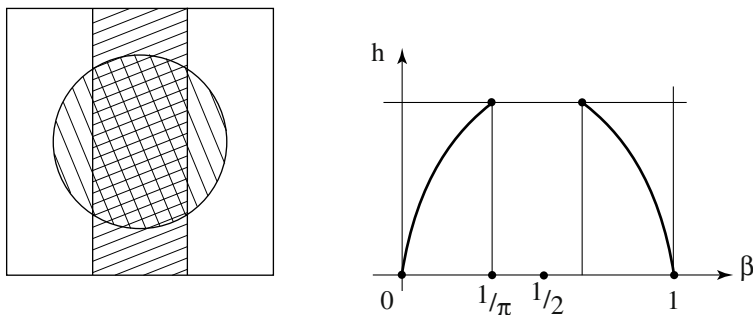


Fig. 7.17. The isoperimetric profile of the flat square torus

For compact hyperbolic space forms, the isoperimetric profile is complicated but theoretically answerable since again we know which boundary to look at.

The recent result of Ritoré & Ros 1992 [1059] solves the problem for real projective 3-space $\mathbb{RP}^3$ (but not for the three dimensional torus). The answer is the “reasonable guess” from figure 7.18 on the facing page.

Except for spheres of any dimension and $\mathbb{RP}^3$, to our knowledge there is not a single example with dimension at least three where the isoperimetric profile is known. See partial results for noncompact manifolds in Hsiang & Hsiang 1989 [740].

This is a surprise for us because we know everything about the metric of each of the $\mathbb{KP}^n$, in particular the geodesic behavior (see §§6.1.6). But the reason is simple. Boundaries will now, starting in dimension three, be hypersurfaces of constant mean curvature in the manifolds under consideration. Of course in flat tori or real projective spaces (or more generally in the $\mathbb{KP}^n$) the metric spheres (boundary of a metric ball) will be of constant mean curvature (because of homogeneity, for example) but there are certainly other constant mean curvature surfaces. For example the tubular neighborhoods of projective subspaces of any dimension in the $\mathbb{KP}^n$ or various cylinders in flat tori. But we do not know if there are other ones and if so who they are. There is no analogue to the result of §§3.4.2 for surfaces in $\mathbb{E}^3$.

Nevertheless, it seems reasonable to guess that the isoperimetric profile for a square flat torus is given by a succession of curves corresponding first to disks, second to tubular neighborhoods of straight lines, third to tubular neighborhoods of planes, etc. ending with tubular neighborhoods of hyperplanes. As we have seen, something like this is true for $\mathbb{RP}^3$.

For all of the $\mathbb{KP}^n$, it also seems reasonable that the isoperimetric profile will be that given by successive tubular neighborhoods of projective subspaces (starting with points). Now if we leave our favourite manifolds to look for theorems about general manifolds, it was discovered in Pansu 1997, 1998 [998, 1000] that the isoperimetric profile can be quite wild; for example, not smooth at the origin. For more see the references in Pansu’s articles and for some

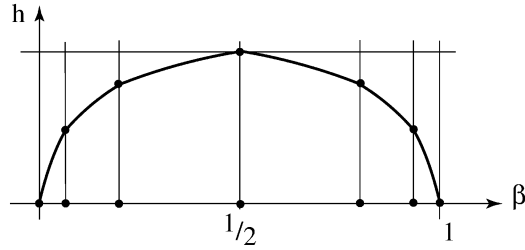


Fig. 7.18. The isoperimetric profile of real projective 3-space $\mathbb{RP}^3$

surfaces of revolution Morgan, Hitchings, & Howards 2000 [938]. In a different direction, the isoperimetric behavior at infinity of noncompact manifolds is basic in various contexts; for instance, group theory and geometry; see Burago & Ivanov 1994 [280].

7.1.2.2 The Gromov–Bérard–Besson–Gallot Bound

Thinking of the proof of Faber–Krahn inequality 1.22 on page 81, assume we want only some bound on the first eigenvalue of the Laplacian, λ_1 (which might not be an optimal bound). Then it is enough to have an isoperimetric bound of the type

$\text{Vol}(\partial D)$ is larger than some function of $\text{Vol}(D)$.

This is what we are going to do for Riemannian manifolds. It is clear that optimal bounds for general Riemannian manifolds are hopeless. But our understanding of control is that we wish to get an estimate from curvature bounds. Recalling the Klingenberg and Cheeger theorems from §§6.5.2, we might expect to need to know the diameter (the volume is already in the profile anyway). After intermediate results due to various authors, Bérard, Besson and Gallot in 1985 got a bound involving only the lower bound of the Ricci curvature and the diameter. The pictures above in §§7.1.2.1 show that one cannot do better. So their bound is optimal given the ingredients. Their explicit functions (which we are not going to give because they are very complicated and not yet having any clear interpretation) will probably be improved some day.

Theorem 114 (Gromov 1980 [633], Bérard–Besson–Gallot 1985[139])
There are three universal functions

$$A_\varepsilon(d, \alpha)$$

for $\varepsilon = 1, 0, -1$ such that for every compact Riemannian manifold M of dimension d with

$$\text{diam}^2 M \cdot \inf \text{Ricci} = (d-1)\varepsilon\alpha^2$$

one has for every β with $0 \leq \beta \leq 1$

$$h(\beta) \geq \frac{1}{\text{diam } M} h_{S^d}(\beta) A_\varepsilon(d, \alpha)$$

where h_{S^d} denotes the isoperimetric profile of the sphere.

We have previously accumulated all of the material for the proof and are prepared for it. The general case is quite involved but the key ideas are in the following simpler case, initiated by Gromov (see Gromov 1999 [633]). The Ricci curvature is positive and we normalize it to have

$$\text{Ricci} \geq d - 1.$$

See also, if needed, IV.H of Gallot, Hulin & Lafontaine 1990 [542]. More precisely, we look only at the positive case and we forget the diameter. This is possible because of Myers' result 62 on page 243. Doing so, we are loosing an improvement of Gromov's inequality, and for the general case we again refer the reader to the original article Bérard, Besson & Gallot 1985 [139]. Before starting, you might like to reread the proof in §§1.6.8 from which we follow the scheme.

In the present case, we want to prove that

$$h_M(b) \geq h_{S^d}(\beta)$$

for every $\beta \in [0, 1]$ and that equality can occur only for the standard sphere, as the proof will show. We pick some β and we use deep results from geometric measure theory (see §§14.7.2) to get a domain $D \subset M$ with volume

$$\text{Vol}(D) = \beta \text{Vol}(M)$$

and with the following properties: (remember the idea on page 64)

1. its boundary ∂D has minimal volume among all domains with same volume as D ,
2. the singular points where ∂D is not locally a manifold are of measure zero in ∂D and
3. when the tangent "object"⁴ to ∂D at some point n is contained in a half space of $T_n M$ then n is a regular point.

These results of *geometric measure theory* are very hard to locate and to find in an handy form; we refer the reader to the various references given in the papers we quoted on the subject, and refer again to Morgan 2000 [937] for a wonderful first introduction to the subject.

We now compute the volume of D as a function of the volume of ∂D as we did in §§1.6.8. For every $m \in D$, let n be a point of ∂D as close as possible to m , and take γ a segment from n to m . Then the first variation formula 6.3 on

⁴ generalizing the tangent plane

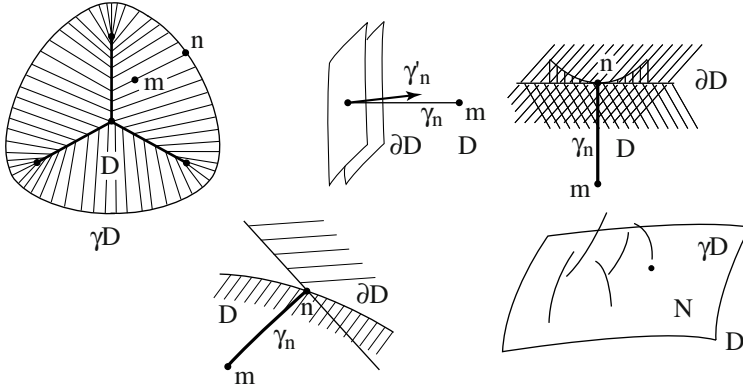


Fig. 7.19. Geometric measure theory in action: Isoperimetric inequalities

page 225 tells us that the tangent object to ∂D at n should stay in the half space of $T_n M$ determined by $\gamma'(n)$ so that n is regular. Denote by N the set of regular points of ∂D .

Starting in the opposite direction, this means that we can recover all of D by starting from all regular points n of ∂D and taking the geodesics γ_n normal to ∂D at n and heading along them inside D . On such a geodesic, we can stop when it ceases to be a shortest path in D . We call this stopping point the *focal value* of n and denote it by $\text{focal}(n)$. As in §§1.6.8 and theorem 86 on page 268 we know that the Heintze–Karcher theorem 111 on page 314 is applicable.

By Milnor’s theorem 109 on page 311 we know that ∂D has constant mean curvature, say η_0 . We apply lemma 101 on page 301 and the Heintze–Karcher theorem 111 to get:

Lemma 115

$$\text{Vol } D = \int_{\partial D} dV_{\partial D}(n) \int_0^{\text{focal}(n)} dt (\cos(t) + \eta_0 \sin(t))^{d-1}$$

Denote by D^* a ball in S^d with

$$\text{Vol } D^* = \beta \text{Vol } S^d .$$

Let r^* be the radius of D^* and η^* be the mean curvature of its boundary. Assume first that

$$\eta_0 < \eta^* .$$

Check that

$$\text{focal}(n) \leq r^*$$

for every $n \in \partial D$ so that the lemma 115 of Heintze and Karcher implies that

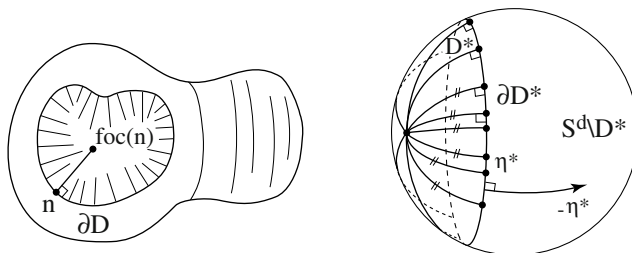


Fig. 7.20. Calculating volume by integrating up to the focal value

$$\text{Vol}(D) \leq \text{Vol}(\partial D) \int_0^{r^*} (\cos(t) + \eta^* \sin(t))^{d-1} dt .$$

When we carry out the same integral for D^* , the last integral will be exactly the same as for D except that it will take place in S^d and with equality everywhere. So the value of this integral is exactly

$$\frac{\text{Vol}(D^*)}{\text{Vol}(\partial D^*)}$$

Q.E.D. We have proven theorem 114 on page 319. If by bad luck $\eta_0 \geq \eta^*$, the amazing trick of Gromov is simply to perform the same computation but with the complementary sets

$$M \setminus D \text{ and } S^d \setminus D^* .$$

This has the effect of switching the orientation of the boundary, so switching the signs of the mean curvatures.

The fun is that is we have equality for only one manifold: the sphere. The above proof will imply knowledge of both D and $M \setminus D$. To prove isometry with the sphere, one just traces back the inequalities, and ends up with a metric given by

$$ds^2 = d\rho^2 + \sin^2 \rho d\sigma^2 .$$

Note 7.1.2.1 Theorem 114 on page 319 will play a significant part in chapter 9 to get lower bounds for every eigenvalue in terms of only the infimum of the Ricci curvature and the diameter. $\blacklozenge$

7.1.2.3 Nonpositive Curvature on Noncompact Manifolds

An unanswered

Question 116 *Is the isoperimetric profile of a complete simply connected Riemannian manifold of nonpositive curvature always dominated by the isoperimetric profile of Euclidean space?*

Such manifolds are never compact, as we know by theorem 72 on page 255 (the von Mangoldt–Hadamard–Cartan theorem). We consider a simply connected manifold M of nonnegative curvature. By the von Mangoldt–Hadamard–Cartan theorem, we know that this manifold is diffeomorphic to $\mathbb{E}^d$. It is conjectured that

Conjecture 117 *For any domain D of M one has the same isoperimetric inequality as for Euclidean space:*

$$\frac{\text{Vol}^d \partial D}{\text{Vol}^{d-1} D} \geq \frac{\text{Vol}^d S^{d-1}}{\text{Vol}^{d-1} B(\mathbb{E}^d, 1)}$$

with equality if and only if the restriction of the metric of M to D is flat and for this flat metric D is a ball.

This was proved for surfaces by André Weil in 1926 using conformal representation and the theory of harmonic functions; it was in fact the first mathematical piece of work of André Weil, answering a question asked during or after a Hadamard seminar at the Collège de France. The lecturer was Paul Lévy, telling the audience about Carleman’s result of 1921. What Carleman had obtained was the desired isoperimetric profile, but only for a minimal surface.⁵ In dimension 4, it was proven by Croke in 1984 using Santalo’s equation 7.11 on page 364. For this demonstration the surprise is that Santalo’s equation gave the right answer only in four dimensions; for all other dimensions it only gives an weaker inequality. For three dimensions, this isoperimetric inequality was recently proven by Kleiner in 1991 [808]; the proof is subtle and uses the fact that hypersurfaces in three dimensions are ordinary Riemannian surfaces.

Unlike two and three dimensional manifolds where one works, so to speak, downstairs (i.e. directly on the manifold itself), the proof for four dimensions requires one to work in the tangent bundle over the manifold, to use the geodesic flow, etc. See §§§7.2.4.2

The main difficulty in proving the above conjecture is that, contrary to the compact case in the beginning of this section, geometric measure theory cannot be applied directly thanks to noncompactness: the optimal domains can “go to infinity.”

Remarkably, Cao & Escobar 2000 [309] proved the desired isoperimetric inequality of our conjecture for three dimensional piecewise linear manifolds. Here, nonpositive curvature means that at the singular points the sum of the measures of the spherical angles is never less than the total volume of the Euclidean unit sphere (of the given dimension). Unhappily, it is an open very interesting question to know if every Riemannian manifold of nonpositive curvature can be nicely approximated by a piecewise linear one (still of nonpositive curvature). The Cao–Escobar result is of course preserved under approximations.

⁵ Recall that minimal surfaces have nonpositive curvature

The natural and fascinating, but very difficult, topic of piecewise flat approximations of Riemannian manifolds is treated in Cheeger, Müller, & Schrader 1984,1986 [350, 351]; see also the lecture Lafontaine 1987 [843].

Note 7.1.2.2 Another important control is that of λ_1 (the first eigenvalue of the Laplacian) with the Cheeger's constant h_c :

$$h_c = \inf \left\{ \text{Vol } D \mid \text{Vol } D \leq \frac{1}{2} \text{Vol } M \right\}$$

which is obtained from the isoperimetric profile by

$$h_c = \inf \left\{ \frac{h(t)}{t} \mid 0 \leq t \leq \frac{1}{2} \right\}.$$

In Cheeger 1970 [329] one finds the inequality

$$\lambda_1 > \frac{1}{4} h_c^2$$

which is optimal by Buser 1978 [291] but never attained by a smooth object. See §§ 9.10.2 on page 409 for more on this. $\blacklozenge$

The beautiful formula in Savo 1996 [1098], or in Savo 1998 [1099], studies the volume of various tubes and then encompasses many results of the above type with a very nice proof. The second derivative of volume with respect to the radius of the tube is computed and linked with the Laplacian. Precisely, let N be any submanifold and u any function. Let $M(r)$ denote the solid tube of radius r around N , and ρ the distance function to N . Then set

$$F(r) = \int_{M(r)} u \, dV_M$$

for the Riemannian measure dV_M . Then

$$-F''(r) = \int_{M(r)} \Delta u \, dV_M + \int_{\rho^{-1}(r)} u \, \Delta u \, dV_{\rho^{-1}(r)}$$

where the measure on the “sphere” $\rho^{-1}(r)$ is the one induced from its Riemannian geometry as a submanifold of M . This formula is valid within the injectivity radius of the tube. Beyond that radius, one has a distribution rather than a function.

Recently, in the spirit of many results in Riemannian geometry (some of them met above), the article of Gallot 1988 [540] succeeded in controlling the isoperimetric profile via integral bounds on the Ricci curvature; see §§§7.2.4.4 for precise statements. There are many corollaries, for example the finiteness theorems of §§12.4.1. and control on the spectrum. Moreover, Gallot also controls the volumes of tubes around hypersurfaces. A key tool as we will see

in the proof of theorem 90 on page 273 or proposition 375 on page 620 would be to control the volume of tubes around geodesics. With only an integral Ricci bound, it is impossible to control the volume of tubes around geodesics; see the Eguchi–Hanson examples in Petersen, Shteingold, & Wei 1996 [1021] for more information in this domain.

The Sobolev inequalities are a basic tool in analysis. They are used in many results quoted in this text.

Theorem 118 (Sergei L. Sobolev) *Take a numerical function*

$$f : M \rightarrow \mathbb{R}$$

on a Riemannian manifold M of dimension d and take integers p and q with

$$\frac{1}{p} + \frac{1}{d} = \frac{1}{q} .$$

Between the function and its gradient one has

$$\|f\|_{L^p} \leq A\|df\|_{L^q} + B\|f\|_{L^q} .$$

In most cases the Riemannian geometer is interested not just in some A and B numbers; he wishes to control them with the curvature, etc. The problems of the optimal A and B are quite different. For the constant B , it is basically the work of Gallot; see the various references to this author above. For the constant A it is the work of Aubin, see for example Aubin 1998 [86]. See the recent references Hebey 1996 [690] and 1999 [691], Hebey & Vaugon 2001 [694] and Druet & Hebey 2000 [463] (where in some cases, sharp estimates of the A term characterize certain Riemannian manifolds). We just mention that in these works the control of the isoperimetric profile is crucial. The isoperimetric profile of a noncompact manifold is of course deeply linked with the geometry of the manifold at infinity. For this important perspective, as well as results, see Pittet 2000 [1032] and the references there.

7.2 Curvature Free Inequalities on Volumes of Cycles

The beauty of this section is its purity: the inequalities we will uncover are curvature independent, so that we completely lack local control—this is an entirely global game. General references are Chavel 1993 [326], Sakai 1996 [1085], two surveys Berger 1993,1996 [165, 168], and in Gromov 1999 [633] chapter 4 and appendix D.

7.2.1 Systolic Inequalities for Curves in Surfaces

7.2.1.1 Loewner, Pu and Blatter–Bavard Theorems

In 1949, Loewner made a wonderful discovery, apparently the very first of its kind. Consider a two dimensional torus T^2 with Riemannian metric g . It

can be abstract (i.e. not embeddable in $\mathbb{E}^3$) but it is not forbidden to be an embedded one.

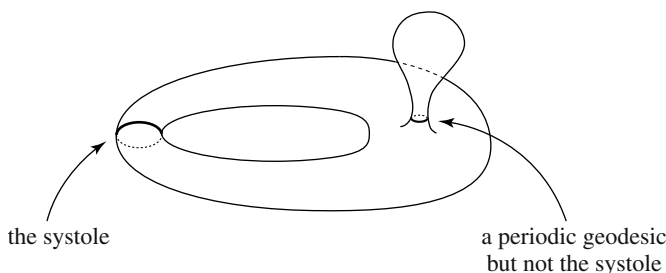


Fig. 7.21. Searching a torus for its systole

We see in the picture in figure 7.21 that there are closed curves on our torus which are not contractible, this is due to the fact that the torus is not simply connected. But beware that the set of closed curves up to homotopy is in general smaller than the fundamental group. This set is the set of conjugacy classes of the fundamental group; see VII.6 of Berger 1965 [151]; the name for a closed curve up to homotopy is “a class of free homotopy.” The torus is an exception to the rule that the classes of free homotopy form a smaller set than the fundamental group, because its fundamental group is Abelian.

Now one might guess that there is, in each free homotopy class, at least one curve of smallest possible positive length. This is not hard to prove. The idea is that the minimum is achieved, even though the space of curves is infinite dimensional, in the present case for various possible reasons. One is the possibility of approximating curves in a compact Riemannian manifold by piecewise geodesics. This reduces the problem to a finite dimensional one (for this idea, basic in Morse theory, see Milnor 1963 [921], and §§§10.3.2). A second reason is the so-called *semicontinuity* of the length of a curve. See for example Choquet 1966 [377], VI.3.11, where there is a proof for quite general metric spaces. Compactness is of course required, as the picture shows. Such a curve will then necessarily be a periodic geodesic, and not merely a geodesic loop. See also 4.12 of Chavel 1993 [326].

We refer to the *systole* of our torus to mean the smallest length of all possible noncontractible curves.⁶ Denote the systole by $\text{Sys}(T^2, g)$. Looking at the picture, a reasonable guess is that if $\text{Sys}(T^2, g)$ is large, then the area of the torus in the metric g cannot be very small.

⁶ The term *systole* comes from physiology (originally from the Greek word for contraction), and refers to the contraction of the heart that occurs when the heart pumps blood into the arteries. The analogy of this periodic behaviour with the periodic behaviour of the geodesics explains the origin of the term in geometry.

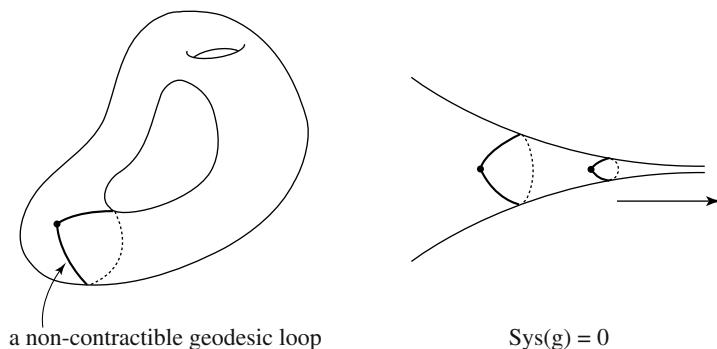


Fig. 7.22. Looking for systoles

Theorem 119 (Loewner 1949) *Any Riemannian metric g on T^2 satisfies*

$$\text{Area}_g(T^2) \geq \frac{\sqrt{3}}{2} \text{Sys}(g)^2 .$$

Moreover equality is achieved precisely when g is flat and corresponds to an equilateral torus (see §§4.3.3).

One can see that inequality as a kind of global isoperimetric inequality: there is no domain with a boundary here, the domain is the whole manifold and the systole plays the role of boundary. Of course we call such an inequality an *isosystolic inequality*. For such an inequality, the word co-isoperimetric could be used, in the sense that here we relate the area to the length of something looking like a boundary, but in the opposite way: instead of squared length bounding a constant multiple of area, we have area bounding a constant multiple of squared length.

The proof is beautiful to be concealed but a little expensive because there is no way to escape to use the conformal representation theorem (which we have already put in its general context in theorem 70 on page 254):

Theorem 120 *Let g be any Riemannian metric on T^2 . Then there exists a flat metric g' on T^2 and a positive function $f : T^2 \rightarrow \mathbb{R}$ such that $g = fg'$.*

Proof. We now give two different proofs of Loewner's theorem. The first is Loewner's original. Being flat, (T^2, g') has a transitive group G of isometries ξ . (The group is identifiable with the torus itself, in some sense, but maintaining a distinction makes the proof clearer.) We calculate the mean value of $f^{1/2}$ under the action of G for its canonical measure $d\xi$ to get

$$(f^*)^{1/2} = \int_{\xi} f(\xi)^{1/2} d\xi .$$

We get a new metric f^*g' . It is easy to check that by the definitions one has

$$\text{Sys}(f^*g') \geq \text{Sys}(g)$$

and (using Schwarz's inequality) that

$$\text{Area}(T^2, f^*g') \leq \text{Area}(T^2, g)$$

so that

$$\frac{\text{Area}(T^2, g)}{\text{Sys}(g)^2} \geq \frac{\text{Area}(T^2, f^*g')}{\text{Sys}(f^*g')^2}.$$

But (if it didn't already occur to you) the function f^* is constant because G acts transitively. So the metric f^*g' is in fact flat. The problem is thus reduced to finding the minimum of the systolic quotient for flat tori. The picture below shows that this is attained exactly for the equilateral one.

The second proof consists in testing the systole on a nice set of curves, large enough to fill up the torus. We fix the flat torus to be given by the two vectors $(0, a)$ and (c, b) in an orthonormal coordinate system (x, y) in the plane, adding moreover that $(0, a)$ is the smallest non-zero vector in the lattice, and that (c, b) is the second smallest one. And, moreover that (c, b) is not a multiple of $(0, a)$ and that

$$c \in \left[-\frac{1}{2}, \frac{1}{2}\right].$$

We compute the sum of the lengths—for the primitive metric fg' —of the horizontal curves and get, by the very definition of the systole:

$$\int_0^b \text{length}(y_{-1}(t)) = \int_0^b \int_0^a f^{1/2} dt dx > b \text{Sys}(g).$$

But

$$\text{Area}(T^2, g) = \int_0^a \int_0^b f dx dt.$$

So we conclude by the Schwarz inequality and looking at the picture in figure 7.23.

Now for some other surfaces. The real projective plane is the simplest. Take any Riemannian metric on it, and look at its Riemannian universal covering, which is a metric on S^2 . Use the conformal representation theorem for the sphere. As an exercise, first compute the systole for the standard real projective plane. After a small trick involving the Möbius group of the standard sphere:

Theorem 121 (Pu 1952 [1043]) *For every Riemannian metric g on $\mathbb{RP}^2$ one has*

$$\text{Area}(\mathbb{RP}^2, g) \geq \frac{2}{\pi} \text{Sys}(\mathbb{RP}^2, g)^2$$

with equality if and only if the metric is the standard one (up to rescaling by a constant factor).

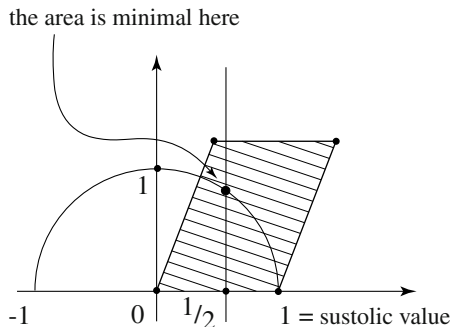


Fig. 7.23. Loewner's theorem

See VIII.12 of Berger 1965 [151] for details. Next comes the Klein bottle. Again we have a complete solution, but the result is not the one which we would naively expect. As an exercise, compute the systolic ratio for the flat Klein bottle. You will have first to know that the flat Klein bottle can only come about from rectangular flat tori.

Theorem 122 *For any metric g on the Klein bottle K ,*

$$\text{Area}(K, g) \geq \frac{2\sqrt{2}}{\pi} \text{Sys}(K, g)^2$$

and if one wants to attain this lower bounded one should take a metric with a singular line obtained by gluing two copies of the Möbius band obtained from the standard sphere as indicated by the picture in figure 7.24 on the next page.

Note 7.2.1.1 The systolic inequality for Klein bottles is largely present in Blatter 1963 [206] and was rediscovered in Bavard 1988 [119]. Why is the optimal Klein bottle not a flat Klein bottle? Because the smallest geodesic in the homotopy class which is of order 2 is isolated. The neighboring curves are turning twice around it. So in the flat case one can reduce the area without changing the systole. The general idea is that there should be continuous families of curves having the systolic length; if not then one can reduce the volume. See below for more on this, and 4.D in Berger 1996 [168], and of course Calabi 1992 [304]. For example, for the equilateral torus there are three such families. The case of $\mathbb{RP}^2$ is exceptional: all of the geodesics are periodic. Look for yourself at these bands for Bavard's Klein bottle. ♦

7.2.1.2 Higher Genus Surfaces

We recall that compact surfaces are completely classified (see figure 3.36 on page 138). The orientable ones are the sphere, the torus and then surfaces with two or more holes; the number of holes γ is called the *genus* and the

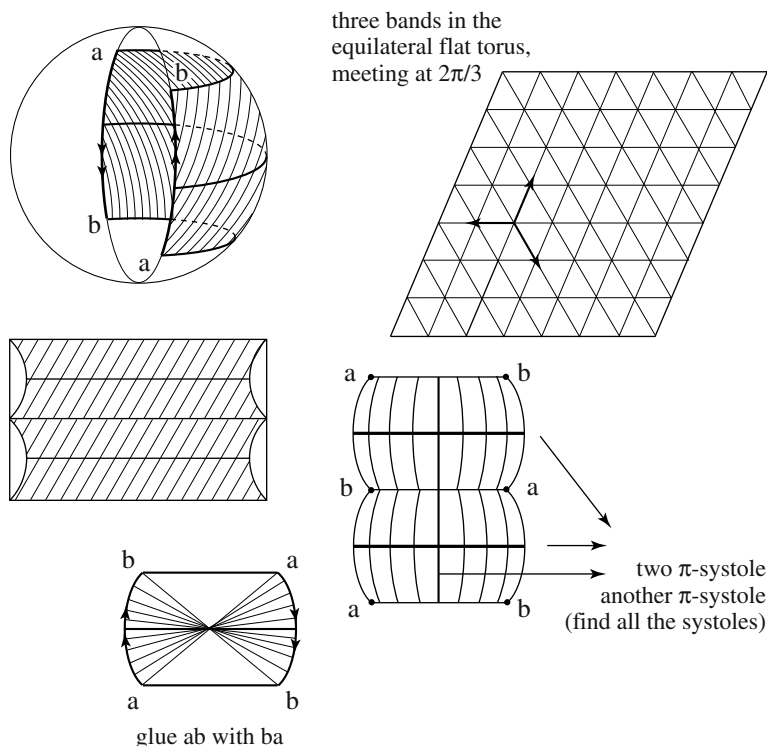


Fig. 7.24. Gluing together a singular Klein bottle with minimal ratio of area to squared systole

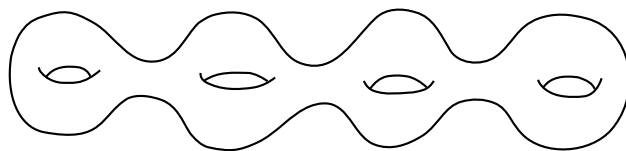
Euler characteristic χ (already met in equation 3.17 on page 138) is equal to $2(1 - \gamma)$. The nonorientable ones are $\mathbb{RP}^2$, the Klein bottle and those of higher genus (obtained by add doughnut holes to $\mathbb{RP}^2$, for example).

Looking at the pictures, it is tempting to think that for surfaces M of genus higher than one, the systolic ratio

$$\frac{\text{Area}(M, g)}{\text{Sys}(M, g)^2}$$

can be given a lower bound and that moreover this bound will grow with the genus.

This was proven recently (in 1983) by Gromov, and has a very interesting history. In various fields of mathematics (algebraic curves, number theory, and the conformal representation theorem 70 on page 254) a very fruitful approach to the study of surfaces of any genus is via the theory of functions of a complex variable. Independently in Accola 1960 [11] and Blatter 1962 [206] complex function theory was used to attack the systolic problem for surfaces of genus higher than one. But counter to the intuition stemming from our pictures they



orientable surface, genus 4

large area ?

Fig. 7.25. An orientable surface of genus 4. Can a surface have large area with small systole, if it has large genus?

could only find a lower bound on the systolic quotient

$$\frac{\text{Area}(M, g)}{\text{Sys}(M, g)^2}$$

which tended toward zero for large genus.⁷ The reason is that they used the conformal representation theorem, always a natural move when studying surfaces. They thereby reduced the question to the isosystolic inequality for flat tori (applied to the Jacobian of the surface M).⁸ See the end of section §§§7.2.2.1 and the remarks at the end of section §§§7.2.2.3. This gave them a bound whose order of magnitude in γ is $1/g^{\gamma/2}$. This is shocking.

This is one of the rare cases where, for studying surfaces, conformal representation (alias complex analysis) is the wrong approach.

Theorem 123 (Hebda & Burago (independently) 1980)

$$\frac{\text{Area}(M, g)}{\text{Sys}(M, g)^2} \geq \frac{1}{2}$$

whatever the nonsimply connected surface and whatever metric g you place on it.

Proof. Let γ be a periodic geodesic of length equal to

$$L = \text{Sys}(M, g)$$

and m any point on γ . We are going to see that

$$\text{Vol } B\left(m, \frac{L}{2}\right) \geq \frac{L^2}{2}.$$

The subtlety is that we cannot say that the injectivity radius is larger than or equal to $L/2$, as one can see on the picture in figure 7.26 on the next page exhibiting “small fingers.” So the simple argument used below in section §§§

⁷ It was not the best possible lower bound, as we will see.

⁸ For more on this, see e.g. section 4.B of Berger 1993 [165].

7.2.4.1 on page 353 is not available. Still we will retain part of it as follows. For any

$$r < L/2$$

we consider the closed metric disks

$$\bar{B}(m, r) .$$

By the very definition of the systole, $\bar{B}(m, r)$ is a topological disk, and in particular its boundary

$$S(m, r) = \partial \bar{B}(m, r)$$

is made up of two curves joining $\gamma(r)$ to $\gamma(-r)$. We cannot have both of these curves of length smaller than $2r$ because then one will get together with the remaining part of γ a closed curve homotopic to γ and smaller than γ , contradicting the fact that γ realizes the systole.

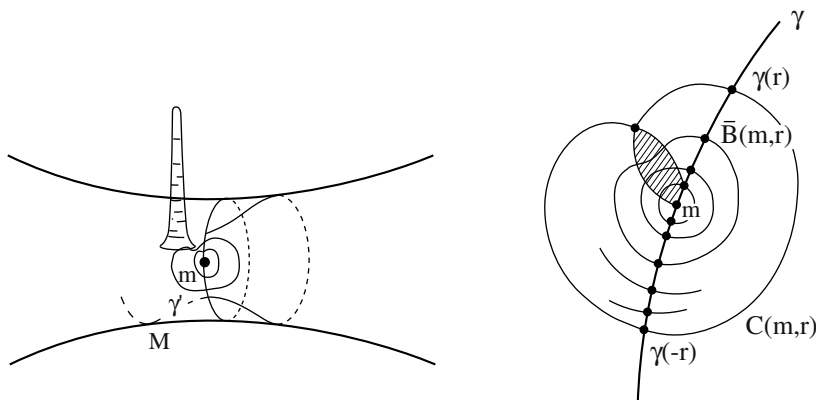


Fig. 7.26. Little fingers

Putting these two curves together, we know that

$$\text{length } S(m, r) \geq 4r$$

and by integration

$$\text{Area } \bar{B}\left(m, \frac{L}{2}\right) = \int_0^{L/2} \text{length } S(m, r) \, dr \geq \int_0^{L/2} 4r \, dr = \frac{L^2}{2} . \quad (7.3)$$

Very soon after, Gromov used the same technique to prove what was expected, namely growth with the genus.

Theorem 124 (Gromov) *For any Riemannian metric on a compact surface M of genus γ*

$$\frac{\text{Area}(M, g)}{\text{Sys}(M, g)^2} \geq a\sqrt{\gamma}$$

for some positive constant a (see also note 7.2.1.2).

Proof. (Sketch) Build up a succession of periodic geodesics γ_i belonging to a basis of the homology of M .⁹ We ask that each γ_i be the shortest in its homology class. The systole is still denoted by L . For any point m on any γ_i we still have

$$\text{Area } B\left(m, \frac{L}{4}\right) \geq \frac{L^2}{8}.$$

Now consider a maximal set of pairwise disjoint balls

$$B\left(m_j, \frac{L}{4}\right)$$

each of which has center m_j belonging to one of the γ_i . We apply the following classical trick, which is no more than the triangle inequality in disguise, but which is incredibly useful in many contexts (see for example chapter 12):

Lemma 125 (The covering trick) *In a metric space, take a system of pairwise disjoint metric balls of a given radius r as large as possible (i.e. so that with larger r some of them will overlap). Then the system of balls obtained from the preceding one by keeping the centers the same, but doubling the radii, covers the whole space.*¹⁰

Remark now that any one of the γ_i needs at least two such balls, and that such a pair of balls cannot be used for any other loop γ_j . So the number N of our balls satisfies

$$\frac{N(N-1)}{2} \geq 2\gamma.$$

Thus

$$N = O(\sqrt{\gamma}).$$

Note 7.2.1.2 Gromov has a much better result on systolic inequalities for surfaces. By deeply refining the above proof, he got a lower bound of the type

$$\frac{\text{Area}(M, g)}{\text{Sys}(M, g)^2} \geq a \frac{\gamma}{\log^2 \gamma}$$

(the constant a being of course universal, i.e. independent of the metric g and the choice of surface M). For the proof, see 6.4.D' of Gromov 1983 [618] (where the wrong factor $\log \gamma$ appears) and 2.C of Gromov 1992 [632], or

⁹ The homology of M is of dimension twice the genus on an orientable surface.

¹⁰ Do not confuse this weak covering trick with the very strong doubling trick of theorem 108 on page 311, which does not apply to general metric spaces.

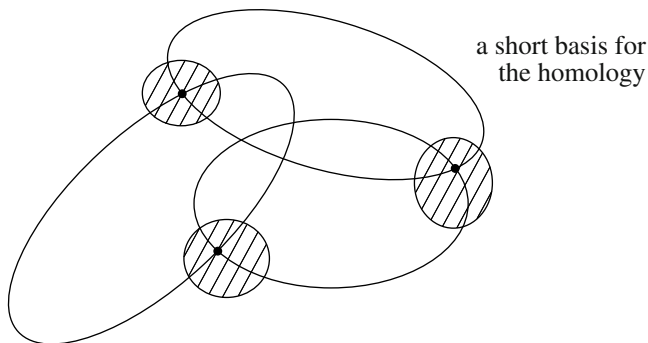


Fig. 7.27. A short basis for the homology

4.C of Berger 1993 [165] for very sketchy idea of the proof. This a very hard result, using Gromov's technique called *diffusion of cycles*. See also VIII. 9. B. 2 and IX. 4. 4. A detailed proof would most welcomed. Gromov can prove that asymptotically the ratio

$$a \frac{\gamma}{\log^2 \gamma}$$

is optimal (unpublished). ♦

Let us consider where the subject currently stands. There are many things to say. Here are the more important and natural. The first remark is that elementary geometry is solving a problem for surfaces that complex function theory was unable to solve. This is exceptional, to our knowledge. We turn now to two natural questions:

Question 126

1. For a given surface, what is the optimal systolic ratio?
2. For which metrics is it attained?

Above we saw that both questions are solved for the torus, the projective plane and the Klein bottle. As far as we know, both questions are open for any other compact surface. There are only partial answers.

The first is in section 5.6 of Gromov 1983 [618]

Theorem 127 *The infimum systolic ratio is attained by some metric, but for a Riemannian metric with singularities.*

The relevant kind of singularity is quite hard to grasp. This is an abstract statement. The value of the lowest systolic ratio and the metric achieving it are not known today. But very hard work was done by Calabi (see below). A result of Jenni 1984 [762] gives the best constant but within the restricted class of metrics of constant curvature (for example, curvature equal to -1) and for hyperelliptic surfaces of genus 2 and 5 only. Recall from §§6.3.2 that

these space forms are very important mathematical objects. A good reference is the recent book Buser 1992 [292]. Also see Bavard 1993 [120] and of course Ratcliffe 1994 [1049].

Note 7.2.1.3 A metric of negative curvature will never achieve the best ratio. The philosophy for this was explained at the end of the preceding section (see figure 7.28). In fact, when the curvature is negative, periodic geodesics are isolated by the second variation formula, and then one can always reduce the volume without changing the systole. But if one decides to work within the world of Riemann surfaces, i.e. Riemannian surfaces of constant curvature, the game is different; see Buser 1992 [292]. ♦

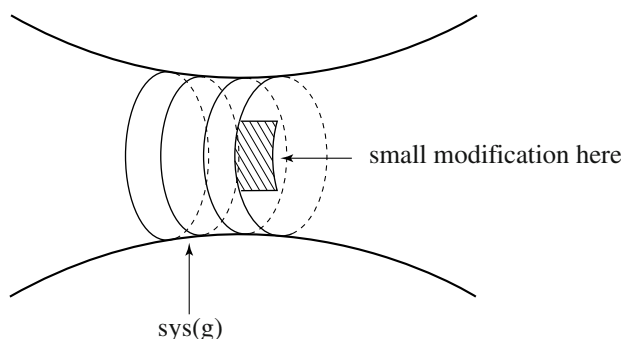


Fig. 7.28. Systoles on negatively curved surfaces

Note 7.2.1.4 We come back now to the general case. In general, when looking for the minimum of some quantity, the natural approach since the appearance of calculus is to look for minima among the cases where the first variation is zero. Then one studies the case at hand directly or computes the second variation, etc. We did that amply for geodesics and for the isoperimetric inequality. But systoles are not accessible to calculus, in some sense because of their nature, or perhaps because we lack the required tools. However Calabi 1992 [304] succeeded in (partly) reducing the problem to studying a partial differential equation. However, analysis of this equation is still in progress. Of course, the first case to look at is genus 2, but it does not seem to be simpler than, say, genus 4. ♦

Let us mention a simply stated plane geometry problem which, according to Calabi 1992 [304], is linked with the above. Consider the first Besicovitch theorem 144 on page 352 applied to a square in the plane: consider all possible Riemannian metrics on the square having the property that the Riemannian distance between any pair of points sitting in opposite sides of the square is always at least one. Then the Riemannian area of this Riemannian square is

at least one, and equality can occur only when the square is the standard Euclidean square. One can compare this with the no-boundary torus case of Loewner.

Conjecture 128 (Calabi) *Suppose we take a hexagon and impose a metric on it so that the side lengths are at least one unit, and so that any pair of points on opposite edges are at least one unit apart. See figure 7.29. He has a conjecture for which we refer again to Calabi 1992 [304]: the extremal metric inside the hexagon looks like “a fried egg;” see figure 7.30 on the facing page. It will have regions with curvature of either sign, with singularities at the junctions.*

We have mentioned Besicovitch’s result because it seems to have a very promising future in Riemannian geometry, since it was revived by Gromov in section 7 of Gromov 1983 [618]. This is essentially studying Riemannian manifolds with boundary; see §§ 14.5.1 on page 676 for more on this topic. One more remark: the story is really only about surfaces and one dimensional systoles, since Besicovitch 1952 [181] proves that for a cylinder, there are metrics with arbitrarily small volume even when both the infima of the length of the curves joining the two boundary disks and the area of disks whose boundary belong to the cylinder itself are larger than one.

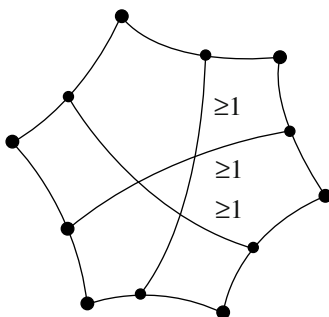


Fig. 7.29. What is the smallest possible area of our hexagon with at least unit distance between opposite sides, and which metric has that area?

7.2.1.3 The Sphere

The notion of systole on the sphere does not make sense at first glance since the systole is the least length of noncontractible curves, and all curves on a sphere are contractible. But the systolic value is achieved by periodic geodesics. So now forget about topology and retain only periodic geodesics. Consider a Riemannian metric g on S^2 and take the smallest periodic geodesic. It is not obvious that there is one. There is, but it is hard to prove and will be

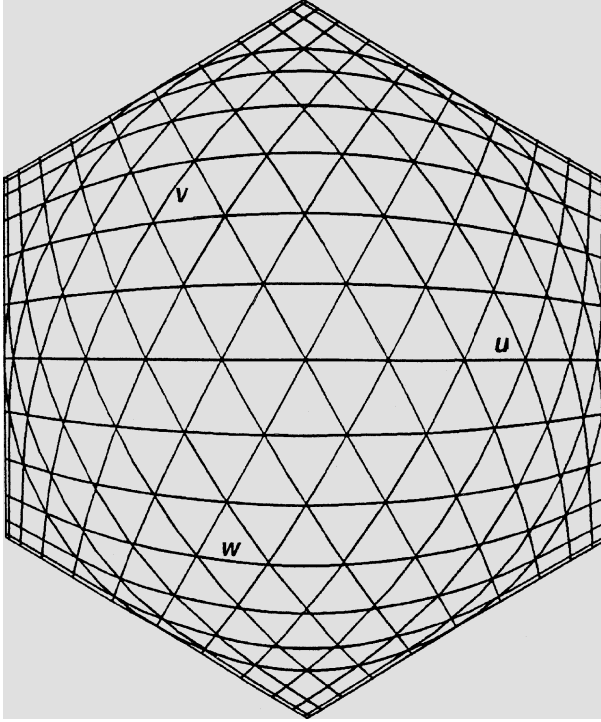


Fig. 7.30. Calabi's fried egg

amply treated in §§10.3.1. Let $\Lambda(S^2, g)$ be the length of that smallest periodic geodesic. You might feel that forces the area $\text{Area}(S^2, g)$ to be large. This question was asked by Gromov in 1980. The conjecture is

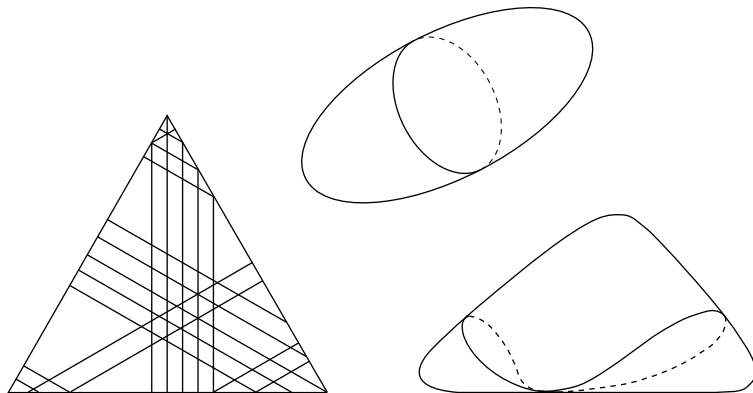
Conjecture 129 *For any Riemannian metric g on S^2*

$$\frac{\text{Area}(S^2, g)}{\Lambda(S^2, g)^2} \geq \frac{1}{2\sqrt{3}}$$

and equality can be achieved only by the singular metric corresponding to a doubly covered equilateral triangle.

Today there is only a partial answer by Croke 1988 [417] with the constant $1/961$. For higher dimensional spheres, one needs to restrict to metrics with some condition on the curvature; see Wilhelm 1997 [1264], Rotman 2000 [1072].

Comparing the Birkhoff closed geodesic and the shortest one, see Nabutovsky & Rotman 2002 [960], Sabourau 2002 [1079, 1080, 1081] and Rotman 2000 [1072].



again three families and meeting at $2\pi/3$

Fig. 7.31. The smallest area that a topological 2-sphere can have with fixed length for the smallest period geodesic occurs for the doubly covered equilateral triangle

7.2.1.4 Homological Systoles

We defined the systole of a surface in terms of the curves which are not homotopic to zero (noncontractible). But in topology, a closed curve defines a homology class, and even a noncontractible curve can have vanishing homology class.

Definition 130 *The homological systole*

$$\text{Sys } H_1(M, g)$$

of a surface M with Riemannian metric g is the lower bound of the lengths of the closed curves whose homology is nonzero.

For the torus, the Klein bottle, or the real projective plane, the two notions coincide.¹¹

But starting at genus two, the homological systole is radically different from the previously discussed free homotopy systole. Figure 7.32 on the facing page presents a famous picture of a noncontractible small closed geodesic which is homologous to zero.

The closed curve drawn there is homological to zero as is visibly obvious since it bounds a subsurface, but (also visibly obvious) it is noncontractible. In Gromov 1992 [632] it is proved, by a nice induction argument starting with the torus and without too much difficulty, that for surfaces of any genus, theorem 124 on page 332 is still valid (asymptotically) for the homological

¹¹ Be careful with the nonorientable case, where you will have to decide whether you prefer to work with the integral or the $\mathbb{Z}_2$ homology.

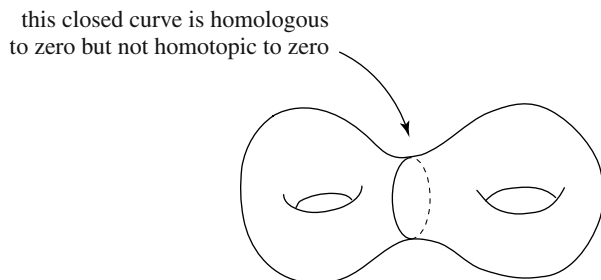


Fig. 7.32. The closed curve drawn is homologous to zero, but not freely homotopic to zero.

systole. The idea of the proof is to use induction, by cutting the surface along a periodic geodesic as in figure 7.33. Close the two pieces with a hemisphere. This will not change the systolic ratios. One piece is a torus and we apply to it Loewner's theorem 119 on page 326 and the induction hypothesis to the second piece. Finally, note that

$$\frac{\gamma}{\log^2 \gamma}$$

behaves essentially additively in γ .

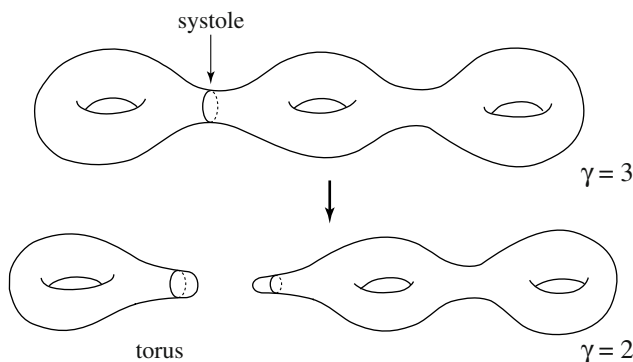


Fig. 7.33. Inductively obtaining asymptotics of homological systoles

This result is surprising since, starting with genus 2, the homology group is much smaller than the homotopy one. The homology of a compact surface is abelian and of finite dimension, while the homotopy group is infinite, and a little like a free group (so with exponential growth). In conclusion one would have been expecting a much smaller ratio for

$$\frac{\text{Area}(M)}{\text{Sys } H_1(M)^2}$$

than for

$$\frac{\text{Area}(M)}{\text{Sys}(M)^2}$$

since many homotopically nontrivial curves are homologically trivial.

7.2.2 Systolic Inequalities for Curves in Higher Dimensional Manifolds

7.2.2.1 The Problem, and Standard Manifolds

Consider now compact Riemannian manifolds of dimension d larger than two and only nonsimply connected ones. For such a manifold M we still have the general notions of systole $\text{Sys}(M)$ and of course the volume $\text{Vol}(M)$. But, thinking of §§7.2.1.4 above, we have to be careful to distinguish between homology and homotopy. We will make a distinction between $\text{Sys } H_1(M)$ and $\text{Sys}(M)$ only when needed and leave the reader in temporary confusion. He will soon discover that the distinction is insignificant (sadly enough).

There is no universal lower bound for

$$\frac{\text{Vol}(M)}{\text{Sys}(M)^{\dim M}}$$

(independent of the manifold M). Consider the manifold

$$M = S^1 \times S^{d-1}.$$

Taking product metrics for which the circle S^1 keeps length at least 1 and the volume of the S^{d-1} factor goes to zero, the systole will be equal to 1 and the volume will be as small as you like.

What was preventing us from employing this technique on a surface? Algebraic topology: on any surface other than the sphere S^2 , the fundamental group (or the first homology group) generates the fundamental class. The systole represents the first homology and the volume the fundamental class. If they are dissociated as in the example above there is of course no hope to get an inequality. If you do not know what the fundamental class of a compact manifold M is, you can consider it as the top dimensional class in the de Rham cohomology group $H_{dR}^{\dim M}(M)$ as described in §§4.2.2.¹²

We have then to restrict the potential manifolds. The simplest examples which come to mind are tori T^d and the real projective spaces $\mathbb{RP}^d$. Also, we can look at space forms of negative curvature type. In those three cases, the first homology generates the fundamental class. No one could prove any systolic inequality for these spaces before Gromov cracked the nut in the very

¹² This is true for orientable manifolds. For unorientable manifolds, one has to work with $\mathbb{Z}_2$ homology.

important article Gromov 1983 [618] which is the basic reference for everything we will present here. See also Gromov 1992 [632].

Before presenting a sketch of Gromov's many new ideas, we look at a naive question which we almost skipped in the two dimensional case. Consider first a flat (locally Euclidean) torus and ask what is the systole and what is the volume? Denote by Λ the lattice yielding this flat torus as presented in §§4.4.3. Geodesics being projections of straight lines, the periodic ones are represented by segments joining the origin to some nonzero point of the lattice Λ and the volume is equal to the determinant of Λ . The systole is then equal to the smallest possible norm for a nonzero element of Λ . Our question is to find what is the best possible ratio for various lattices in $\mathbb{E}^d$ of the determinant to the d^{th} power of the smallest nonzero element of the lattice.

In two dimensions, the best ratio was depicted in figure 7.22 on page 327. For a general dimension d this is a famous problem in the field of mathematics called *geometric number theory*. Compactness shows quite easily that there is a optimal lower bound and that it is achieved by some lattices, called *critical* lattices. They are completely known in vector spaces of all dimensions up to and including eight dimensions. After eight dimensions, neither the best ratio nor the lattices achieving it are explicitly known, but extremely good asymptotic values are known. We refer the reader to Berger 1993 [165], Gromov 1992 [632] and the various references there. We just want to note that the above bound is extremely small in d ; its order of magnitude is

$$\frac{1}{d^{d/2}}.$$

The best references in our opinion are Gruber & Lekkerkerker 1987 [660] and Conway & Sloane 1999 [403]. Erdős, Gruber & Hammer 1989 [491] could be also useful.

An even more special case of flat tori is of great interest in algebraic geometry and number theory, namely we consider the so-called *Jacobian varieties* of Riemann surfaces. They are flat tori of (real) dimension 2γ if the genus of the surface is γ . Jacobians have a natural (flat) metric depending on the conformal structure on the given surface. It is a long standing problem to characterize Jacobians among flat tori (or equivalently, to characterize the lattices in $\mathbb{C}^\gamma$ which yield Jacobians). There are some very algebraic and complicated solutions of that problem; see Beauville 1987 [123]. Buser & Sarnak 1994 [297] proved that the order of magnitude of the systole is very different from the general case where the order of the systole which can go up to $\sqrt{\gamma}$, namely the systole has maximal order $\log \gamma$. See on page 352 for a geometric characterization of Jacobians using 2-systoles.

In our beloved $\mathbb{K}\mathbb{P}^n$, the only case to consider is the real projective space (with its canonical metric) for which things are trivial: there is only one metric on projective space of constant curvature and since we know the geodesics and the total volume we are done.

7.2.2.2 Filling Volume and Filling Radius

Gromov proved the existence of a positive lower bound for the systolic ratio

$$\frac{\text{Vol}(M)}{\text{Sys}_d(M)}$$

for any compact Riemannian manifold M whenever M has sufficient 1-dimensional topology to generate its fundamental class. The exact condition will be expressed in §§7.2.2.3. Before describing that condition, we will now introduce the concepts needed for the proof. These concepts have already much geometric interest and raise many open questions.

The difficulty in the topic—at least with current knowledge—is that one cannot avoid working in infinite dimensional vector spaces. The starting point is to construct the map

Definition 131

$$f : M \rightarrow L^\infty(M)$$

where $L^\infty(M)$ is the space of all bounded functions on M endowed with the sup norm and where the map f is given by the various distance functions

$$f : m \mapsto d(m, \cdot)$$

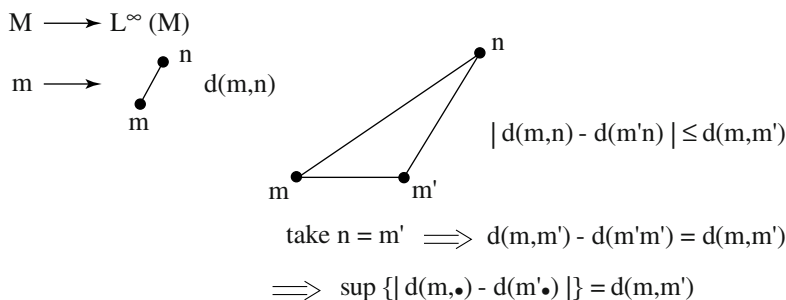


Fig. 7.34. The map taking a point $m \in M$ to the distance function $d(m, \cdot) \in L^\infty(M)$

For the two metric spaces M and $L^\infty(M)$, the map f is an isometry.

Definition 132 The filling radius of a compact Riemannian manifold M is the smallest positive number ε for which the image $f(M)$ in $L^\infty(M)$ bounds in its ε tubular neighborhood.¹³

¹³ The term “to bound” comes from algebraic topology and means that there is a cycle of one higher dimension whose boundary is the object under consideration.

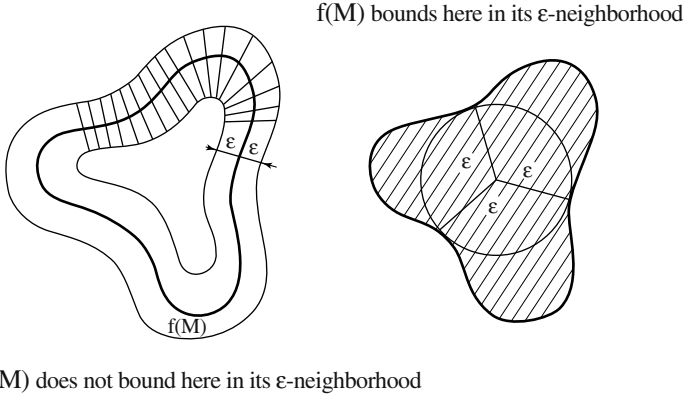


Fig. 7.35. (a) $f(M)$ does not bound in its ϵ neighborhood. (b) $f(M)$ bounds here in its ϵ neighborhood.

This filling radius is a geometric invariant which at first glance appears very simple. In fact its value is known only for the standard spheres S^d , the standard projective spaces $\mathbb{RP}^d$ and the complex projective plane $\mathbb{CP}^2$, but is still unknown for the remaining $\mathbb{KP}^n$ as well as for any other Riemannian manifolds (in particular, any of the above manifolds with nonstandard metrics); see Katz 1983,1991 [792, 793] and Wilhelm 1992 [1261]. For the filling radius of the sphere, see Sabourau 2002 [1079, 1080, 1081].

Definition 133 *The filling volume of a compact Riemannian manifold M of dimension d is the infimum of the volumes of the $d + 1$ dimensional submanifolds in $L^\infty(M)$ whose boundary is $f(M)$.*

Today there is not a single manifold whose filling volume is known, not even the circle (for which Gromov conjectures the value is 2π). These two invariants are promised a future in Riemannian geometry. For example the filling radius is used in Greene & Petersen 1992 [592] to improve the recent finiteness theorems mentioned in §§ 12.4.1 on page 614. But some inequalities (due to Gromov of course) are valid for any compact manifold; namely

Theorem 134 *There are two positive constants $a(d)$ and $b(d)$ such that for every compact d dimensional Riemannian manifold M*

$$\begin{aligned} \text{Vol}(M)^{d+1} &> a(d) (\text{Filling volume of } M)^d \\ \text{Filling volume of } M &> b(d) (\text{Filling radius of } M)^{d+1} \end{aligned}$$

The proofs are hard, especially since one is working in infinite dimension. But as seen in ordinary space they are quite reasonable. The first inequality is classical in the standard Euclidean setting, say of dimension N . You have a compact d dimensional submanifold M in $\mathbb{R}^N$ and you span it with a minimal

submanifold H of dimension $d + 1$ with the boundary $\partial H = M$ equal to your original manifold. Then there is an isoperimetric inequality in Michael & Simon 1973 [918] to the effect that

$$\frac{\text{Vol}(M)^{d+1}}{\text{Vol}(H)^d} > a(d)$$

for some universal constant $a(d)$ (independent of M and H).

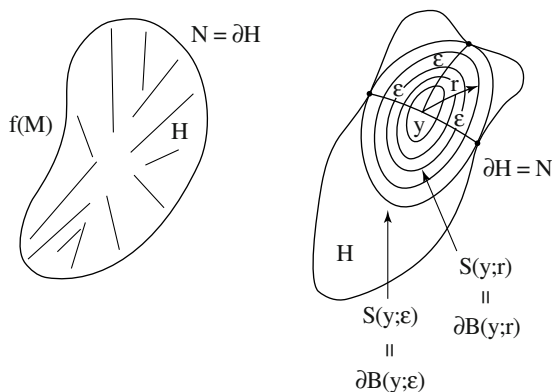


Fig. 7.36. There is a isoperimetric inequality of classical character for minimal submanifolds and their boundaries

We will continue to imagine this picture in a finite dimensional Euclidean space $\mathbb{R}^N$ of M as the boundary

$$\partial H = M$$

of a $d + 1$ dimensional minimal submanifold H of $\mathbb{R}^N$. (Of course, things can be more subtle in infinite dimensions.) Now we will use the definition of the filling radius ε . This means exactly that there is some point $y \in H$ such that the metric ball $B(y, \varepsilon)$ is filled up nicely with the spheres $\partial B(y, r)$ for every $r < \varepsilon$. Compare this with the proof in theorem 123 on page 331. Using this and the minimality we see that volume should grow at least as quickly as in the flat case.

Note 7.2.2.1 Filling radius and filling volume have been introduced here, not only because they are basic ingredients in Gromov's proof, but because it seems highly probable that they will play an important role in the future of Riemannian geometry. $\blacklozenge$

7.2.2.3 Gromov's Theorem and Sketch of the Proof

First we state the technical formulation of the heuristic notion of the first dimensional homology generating the fundamental class.

Definition 135 *A topological space is said to be aspherical if except for dimension one all the homotopy groups vanish.*

Definition 136 *A compact Riemannian manifold M is said to be essential if there is some map*

$$g : M \rightarrow X$$

into some aspherical space X such that the induced map

$$g_* : H_*(M) \rightarrow H_*(X)$$

at the homology level maps the fundamental class of M into a nonzero element:

$$g_*[M] \neq 0.$$

Theorem 137 (Gromov 1983 [618]) *There is a positive constant $c(d)$ such that for every essential compact d dimensional Riemannian manifold M one has*

$$\frac{\text{Vol}(M)}{\text{Sys}_1(M)^d} > c(d).$$

(where $\text{Sys}_1(M)$ is the homotopic 1-systole).

Using theorem 134 on page 343 the proof is finished with

Theorem 138 *For any essential compact Riemannian manifold M*

$$\text{Filling radius}(M) \geq \frac{1}{6} \text{Sys}_1(M).$$

This is the key point in the proof and it goes by contradiction assuming that

$$\text{Filling radius} = r < \frac{1}{6} \text{Sys}_1(M).$$

We first identify M with its isometric image $f(M)$ inside $L^\infty(M)$. Then we know that M is the boundary of some cycle H which is contained in the r neighborhood of M , since r is the filling radius of M . This means that the inclusion map acts on homology

$$\iota_* : H_d(M) \rightarrow H_d(K)$$

trivially on the fundamental class of M :

$$\iota_*[M] = 0.$$

Let

$$\eta : M \rightarrow X$$

be a continuous map into an aspherical space X which is not trivial on the fundamental class of M at the d^{th} dimensional homology:

$$\begin{aligned}\eta_* : H_d(M) &\rightarrow H_d(X) \\ [M] &\mapsto \eta_*[M] \neq 0.\end{aligned}$$

We claim that under the assumption above we can extend η to a map

$$\eta : H \rightarrow X.$$

This will be the desired contradiction, since M is a boundary in H .

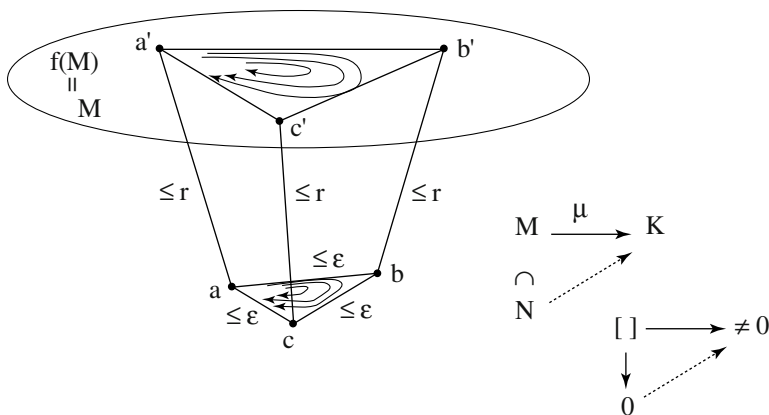


Fig. 7.37. Extending a continuous map from M to a higher dimensional cycle containing M

We note that we have only to carry out such an extension at the two dimensional level since X is aspherical. To do this we triangulate H (including M) in such a way that every edge is of length smaller than ε with

$$3\varepsilon + 6r < \text{Sys}_1(M).$$

To extend η to H , we associate to each vertex $a \in H$ in the triangulation another vertex $a' \in M$ with distance

$$d(a, a') < r.$$

Let $\{a, b, c\}$ be any triangle of the triangulation and $\{a', b', c'\}$ the corresponding triangle in M . The segments which are the sides of $\{a', b', c'\}$ are geodesics in M since we have an isometric embedding. We draw the sides of $\{a, b, c\}$ and this gives us an extension from the original η up to the 1-skeleton. Now we remark that by our construction and the triangle inequality

$$\text{perimeter of } \{a', b', c'\} \leq \text{perimeter of } \{a, b, c\} + 6r < \text{Sys}_1(M).$$

By the very definition of the systole, the perimeter of $\{a', b', c'\}$ is contractible in M . In order to extend η to the interior of $\{a, b, c\}$, we have only to make a point to point correspondence with a contraction in M of the perimeter of $\{a', b', c'\}$. This yields the desired extension of η to the 2-skeleton.

Note 7.2.2.2 The above theorem applies to every torus, to every negative curvature type space form (see §§6.3.2) and also to the projective spaces $\mathbb{RP}^d$. To see this you need to know that the infinite dimensional real projective space $\mathbb{RP}^\infty$ is aspherical. $\blacklozenge$

We are left with three questions. First:

Question 139 *What is the optimal value and if “extremal metrics” exist who are they?*

For this, besides the “Filling paper” of Gromov 1983 [618], see Calabi 1992 [304] and Gromov 1992 [632]. But in dimensions three and higher, there is to our knowledge no existence theorem of an optimal systolic metric (possibly with singularities) on any type of manifold. Nor is any optimal ratio known. It is not clear today if it is reasonable to conjecture that for the tori T^d and the projective spaces $\mathbb{RP}^d$ the best ratio is attained for flat and standard metrics and only for them (see §§7.2.2.1). The first cases to look at would be T^3 and $\mathbb{RP}^3$. A classical idea is to sit at the standard metric g for $\mathbb{RP}^3$ or a flat one g for T^3 which is optimal among flat metrics and try to compute a variation of the systolic ratio

$$\frac{\text{Vol}(M, g)}{\text{Sys}_3(M, g)}.$$

This is difficult because the systole is not directly accessible to calculus: see question 126 on page 334. The volume is accessible to calculus. One still can show that the first variation is always zero but the second variation seems hard to work with. See Besse 1978 [182] 5.90 for a result in this spirit.

The second question:

Question 140 *What characterizes compact manifolds for which we have such a universal inequality for*

$$\frac{\text{Vol}(M, g)}{\text{Sys}_1(M, g)^{\dim M}}?$$

This is studied in Babenko 1992 [91] for the homotopical

$$\text{Sys}_1(M)$$

associated to the fundamental group. Such a result can be seen as a “systolic” characterization of a topological property.

The third question concerns the homological systole:

Question 141 *What characterizes compact manifolds for which we have such an universal inequality for*

$$\frac{\text{Vol}(M, g)}{\text{Sys } H_1(M, g)^{\dim M}}?$$

Things are completely solved positively in Babenko 1992 [91] (but not everything figures explicitly in the text, and be careful about the nonorientable case).

7.2.3 Higher Dimensional Systoles: Systolic Freedom Almost Everywhere

Everything in the last section concerned closed curves in a Riemannian manifold. But there is no reason not to consider closed submanifolds of any dimension. For simplicity, we will stick from now on with homology (no longer homotopy). References are Berger 1993 [165], Gromov 1992 [632], chapter 4 and appendix D of Gromov 1999 [633] and references there. Interesting examples of systolic freedom occur in Freedman 2000 [521], where they are related to quantum computing; this relationship is still being developed. See Nabutovsky & Rotman 2002 [960], Katz, Kreck & Suciu 2002 [797], Katz & Suciu 2001 [799], Bangert & Katz 2002 [113], and Katz 2002 [796] for relations with calibrations (see §§14.2.1.3).

We define the k dimensional systole

$$\text{Sys}_k(M, g)$$

of a Riemannian manifold M with Riemannian metric g as the lower bound of the volume of the closed k dimensional submanifolds of M which are not homologous to zero. We will ask

Question 142 *Is there, at least for some compact d dimensional manifolds M , a constant $a(M)$ and an inequality*

$$\frac{\text{Vol}(M, g)^k}{\text{Sys}_k(M, g)^d} \geq a(M)$$

for any Riemannian metric g on M ?

We might ask this for example for the $\mathbb{K}\mathbb{P}^n$ or various products like

$$S^k \times \cdots \times S^k$$

$$M \times N$$

for

$$\frac{\text{Vol}(M \times N)}{\text{Sys}_p(M) \text{Sys}_q(N)}$$

etc. One will find in the two above references the state of affairs today. But we will now be extremely brief for the following reason. It might well be the answer to question 142 is *no* for every compact manifold and every $k \geq 2$.

For higher dimensional systoles, negative results started to appear in Gromov 1992 [632]. They were intermediate results. Gromov's example of $(1, 3)$

softness on $S^1 \times S^3$ is elementary and consists simply in gluing on a copy of $[0, 1] \times S^3$ after twisting enough with the Hopf fibration. We have now a extremely large category of negative examples in Bérard Bergery & Katz 1994 [146], Babenko & Katz 1998 [93] and Babenko, Katz & Suciu 1998 [94]; also see Katz's appendix in Gromov 1999 [633] and Pittet 1997 [1031]. One says that a manifold M^d is *systolically* $(k, d - k)$ *soft* (or *systolically* $(k, d - k)$ *free*) if the infimum of the quotients

$$\frac{\text{Vol}(M, g)}{\text{Sys}_k(M, g) \text{Sys}_{d-k}(M, g)}$$

among all metrics g on M^d is zero. The above authors proved softness in the following cases:

1. For any orientable $(k - 1)$ -connected d dimensional manifold with

$$d \geq 3, \quad k < \frac{d}{2}, \quad \text{and } k \text{ not a multiple of } 4,$$

there is $(k, d - k)$ softness.

2. For simply connected manifolds of even dimension $M^{d=2n}$ with $d \geq 6$, provided $H_n(M)$ is torsion free, there is (n, n) -softness.

The $S^k \times S^k$ (for $k \geq 3$) and $\mathbb{H}\mathbb{P}^2$ are startling examples. The $(4, 4)$ -softness of $\mathbb{H}\mathbb{P}^2$ is most surprising, since the projective lines fill up the whole space in the most geometric possible fashion and should have been “sniffing around enough” to prevent softness. The “freedom” can be very large, for example on $S^3 \times S^3$ one can even restrict attention to only homogeneous metrics and still see the freedom. It is not clear today if the various topological restrictions are really necessary as well as the dimension ones. In fact there is no known example of a hard inequality as soon as the involved systoles are in dimension 2 or more. The dimension 4 case was solved in Katz 1998 [795]. Even earlier, Katz had shown that $S^2 \times S^2$ and $\mathbb{C}\mathbb{P}^2$ are systolically soft. Of course the case of $\mathbb{C}\mathbb{P}^2$ is even more surprising, just as for all $\mathbb{K}\mathbb{P}^n$ but especially because Gromov proved hardness for metrics close to the canonical metric, announced in Gromov 1985 [619]. The recent Katz & Suciu 1999 [798] and Babenko 2000 [92] completely finishes the proof of middle dimensional freedom on even dimensional manifolds.

All the above results are very geometric. We will give a flavor of the ideas and refer to the Appendix D of Gromov 1999 [633] for more details. We start with Gromov's initial example:

Proposition 143 *On the manifold $M = S^1 \times S^3$ there are Riemannian metrics g whose quotient*

$$\frac{\text{Vol}(M, g)}{\text{Sys}_1(M, g) \text{Sys}_3(M, g)}$$

is arbitrarily small.

Proof. We will play with a real number R which will soon start to get larger and larger. Consider M the product of the sphere $S^3(R)$ of radius R with the interval $[0, 1/R]$ with the two boundary spheres identified

$$S^3(R) \times \{0\} \quad \text{and} \quad S^3(R) \times \{1/R\}$$

under the “Hopf rotation” map

$$S^3(R) \rightarrow S^3(R).$$

A Hopf rotation is the following map: we choose some Hopf fibration on $S^3(R)$ by great circles and then push any point by a unit distance along the fiber through it. Then the total volume of the resulting Riemannian manifold (a local product) is

$$2\pi^2 R^3 \cdot \frac{1}{R} = 2\pi^2 R^2.$$

Controlling the systoles is easy:

$$\text{Sys}_3(M) = 2\pi^2 R^3$$

and

$$\text{Sys}_1(M) = \sqrt{1 + \frac{1}{R^2}}.$$

$$\frac{\text{Vol}(M)}{\text{Sys}_3(M) \text{Sys}_1(M)} \sim O\left(\frac{1}{R}\right) \rightarrow 0 \text{ as } R \rightarrow \infty$$

This counterexample metric is not very wild, since it is homogeneous.

To determine systoles on some more general manifolds, one performs surgery in such a manner that one can control the systole and the volume, employing quite a few subtle results of algebraic topology. Technically, one also uses calibrated geometry; see §§14.2.1. Using the concept of calibration, one can relate homological systoles to differential forms, which is not the case in general. When there are topological restrictions, it is because they are needed to be able to employ various algebraic topology results.

Note 7.2.3.1 (Stable systoles) There are results similar to those we have already studied, but for the so-called *stable systoles*. For details see chapter 4 of Gromov 1999 [633] or section 8 of Berger 1993 [165]. We briefly explain the theory of stable systoles. The story here is really in the realm of homology. As an introduction, let us consider the case of some Riemannian surface (you can think of a torus for example). Let γ be the smallest homologically nontrivial periodic geodesic. Travel along it twice; call this 2γ . Can you find some new curve σ which is homotopic to 2γ but of smaller length? An old theorem of Morse says that this is impossible, the reason being that the new curve will

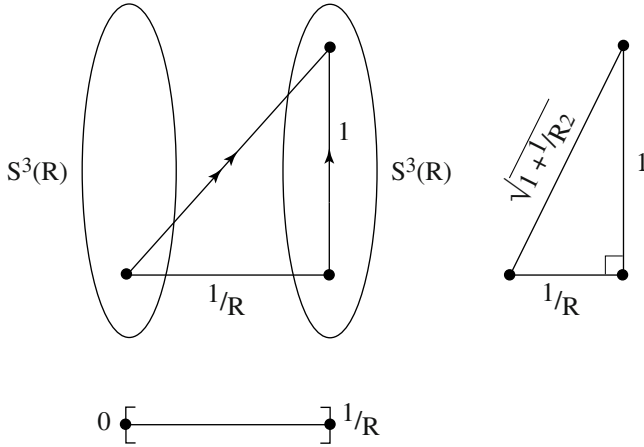


Fig. 7.38. The construction of a homotopically soft 4-manifold

necessarily intersect itself and then will not be a shortest curve by the first variation formula. But in higher dimension, there is room enough to avoid self-intersections. This was already mentioned in Hedlund 1932 [695]. But if we write

$$\text{length } \alpha$$

for the infimum of the lengths of all closed curves in homology class $\alpha \in H_1(M)$, then nonetheless one can prove that there is a positive limit for the real number

$$\text{Stab Sys } H_1(M) = \lim_{n \rightarrow \infty} \frac{\inf \{ \text{length}(n\alpha) \mid \alpha \in H_1(M) \}}{n}$$

which number which we call the *stable 1-systole* of the compact Riemannian manifold M .

One can define the same way the stable systole of any dimension k in any compact manifold M , denoted $\text{Stab Sys } H_k(M)$.

It turns out that stable systoles enjoy the inequality one would naturally expect, e.g. for the $\mathbb{CP}^n$ and for products of manifolds. The reason at the heart of this is that the stable systole can be related to various norms of exterior differential forms on a Riemannian manifold, which is not true for standard systoles as is clear from the counterexamples above. One has at ones disposal the exterior product, Stokes' formula, etc. This is to be found only in chapter, 4, section D of Gromov 1999 [633]. In particular, for $\mathbb{CP}^n$ we see an inequality with the optimal constant

$$\frac{\text{Vol}(\mathbb{CP}^n, g)}{(\text{Stab Sys } H_2(\mathbb{CP}^n, g))^n} \geq \frac{\text{Vol}(\mathbb{CP}^n, \text{Fubini-Study})}{(\text{Vol}(\mathbb{CP}^1, \text{Fubini-Study}))^n}$$

for any metric g on $\mathbb{CP}^n$. For products, one still finds inequalities but, at least today, with poor constants:

- For any Riemannian metric g on $\prod^n S^k$

$$\frac{\text{Vol}(\prod^n S^k, g)}{\text{Stab Sys } H_k(\prod^n S^k, g)^n} > c > 0$$

- For any Riemannian metric g on a product of compact manifolds $M^p \times N^q$

$$\frac{\text{Vol}(M \times N, g)}{\text{Stab Sys } H_p(M, g) \text{Stab Sys } H_q(N, g)} > c > 0$$

◆

Note 7.2.3.2 k -systoles with $k > 1$ are interesting objects. In 2.E of Gromov 1992 [632], Gromov uses some 2-systoles to give a purely geometrical characterization of Jacobians of curves among flat tori, already met for the 1-systole in §§§7.2.2.1. ◆

Note 7.2.3.3 For simply connected manifolds, it makes sense as in §§§7.2.1.3 to consider the volume and the length of the smallest periodic geodesic. Today one can get results if one adds some curvature conditions, see Rotman 2000 [1072]. ◆

Note 7.2.3.4 (Besicovitch's results 1952 [181]) We imagine that the reader is surprised by the theory of systoles, which one can sketch as: for curves, one has very good positive results, but as soon as one works with submanifolds of dimension two or more, one has softness, i.e. rather weak results. In fact this is not surprising, in view of two results of Besicovitch for Riemannian manifolds with boundary:

Theorem 144 (Besicovitch 1952 [181]) *Take the standard topological cube*

$$C = [0, 1]^d$$

but put inside it any Riemannian metric. If L_i denotes the infimum of the length of any curve having its ends in the opposite faces

$$E_i = \{x_i = 0\} \cap C \quad F_i = \{x_i = 1\} \cap C$$

for $i = 1, \dots, d$ then the total volume $\text{Vol } C$ satisfies

$$\text{Vol } C \geq \prod_{i=1}^d L_i$$

with equality if and only if the metric we have imposed on C is the standard Euclidean cube metric.

Theorem 145 (Besicovitch 1952 [181]) *Consider a cylinder*

$$C = D \times [0, 1]$$

where D is a disk. Put any Riemannian metric inside and define L to be the infimum of the length of the curves with one end in the disk

$$D \times \{0\}$$

and the other in the disk

$$D \times \{1\}$$

and denote by S the infimum of the area of surfaces in C whose boundary lies in the surface

$$S^1 \times [0, 1].$$

Then the ratio

$$\frac{\text{Vol } C}{LS}$$

can be made arbitrarily small (softness).

For more see Gromov 1983 [618].

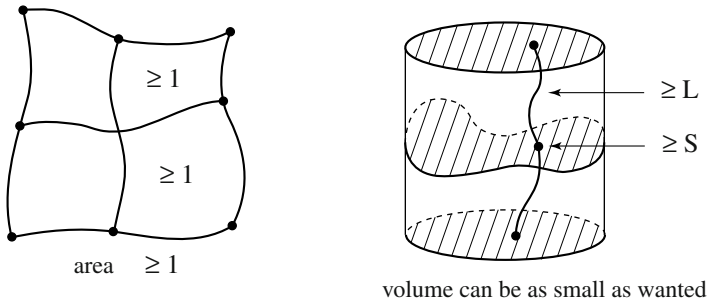


Fig. 7.39. (a) Area ≥ 1 (b) Volume can be as small as desired

7.2.4 Embolic Inequalities

7.2.4.1 Introduction, Questions and Answers

To our knowledge, the content of the present section is treated in book form only in chapters 6 and 7 of Chavel 1993 [326], and not completely even there. One might also consult Sakai 1996 [1085] chapter VI.

Looking at the above two pictures and also remembering the dichotomy of Klingenberg's theorem 89 on page 272, one is tempted to say that forcing the

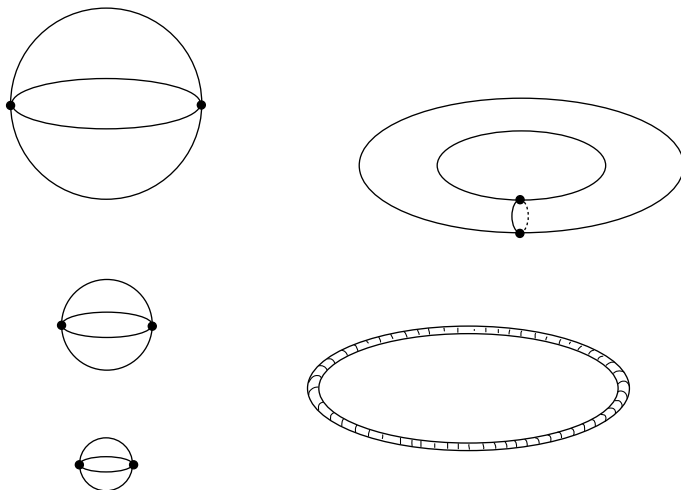


Fig. 7.40. Volume and injectivity radius

volume to be small will force the injectivity radius to be small. For example think of a flat torus or of a sphere.

This is easily seen on any surface; we apply the technique of equation 7.3 on page 332 as follows. Pick up any point $m \in M$ and look at the ball $B(m, \iota/2)$ where $\iota = \text{Inj}(M, g)$. That ball is nicely covered by the exponential map, by the definition of injectivity radius. For any $t < \iota/2$ and any geodesic γ through $m = \gamma(0)$ the two points $\gamma(r)$ and $\gamma(-r)$ cannot be joined by a curve of length smaller than $2r$, once again by the definition of the injectivity radius. In conclusion, the circle $C(m, r) = \partial B(m, r)$ is always made up of two pieces, each of length not smaller than $2r$. By integration one finds as in equation 7.3 on page 332

$$\text{Vol } B(m, \iota/2) \geq \frac{\iota^2}{8}.$$

Since one can cover M with two such balls we know that

$$\text{Vol}(M, g) \geq \frac{\text{Inj}(M, g)^2}{4} \quad (7.4)$$

This technique is very primitive. Indeed one never has equality (proof is left to the reader). Moreover it does not extend to higher dimension, since in higher dimensional manifolds, the boundary of a ball $B(m, r)$ is a sphere $S(m, r) = \partial B(m, r)$ which is no longer a curve and its area is almost impossible to control, at least by any simple mechanism.

We still can ask some natural questions. The first two stem directly from what we have just considered.

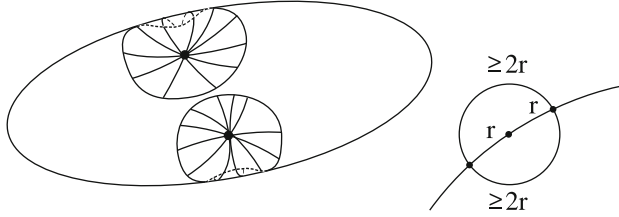


Fig. 7.41. Relating the injectivity radius of a surface to its volume

Question 146 *Is there a positive constant $a(d)$ such that for every compact Riemannian manifold M of dimension d , any point $m \in M$ and any $r \leq \text{Inj}(M)/2$, the volumes of balls satisfy*

$$\text{Vol } B(m, r) \geq a(d)r^d?$$

If so, what is the largest possible choice of $a(d)$?

Question 147 *Is there a positive constant $b(d)$ such that for every compact Riemannian manifold of dimension d the volume of the manifold satisfies*

$$\text{Vol}(M) \geq b(d)\text{Inj}(M)^d?$$

If so, what is the largest possible choice of $b(d)$?

Today there are some pleasing results, but not all of these questions answered. Here is what we know:

Theorem 148 (Berger 1980 [160]) *Question 147 has a positive answer. The largest possible constant is*

$$b(d) = \frac{\sigma(d)}{\pi^d}$$

(recall that $\sigma(d)$ denotes the volume of the standard sphere and that π is its injectivity radius). Moreover equality is attained only for standard round spheres.

Theorem 149 (Croke 1980 [411]) *Question 146 has a positive answer with the (not optimal) bound*

$$a(d) = \frac{2^{d-1}\sigma(d-1)^d}{d^d\sigma(d)^{d-1}}.$$

Theorem 150 (Croke 1984 [415]) *Question 146 has a positive answer with an optimal bound “on average.” Namely, for every $r < \text{Inj}(M)$,*

$$\frac{1}{\text{Vol}(M)} \int_M \text{Vol } B(m, r) dV_M(m) \geq \sigma(d) \frac{r^d}{\pi^d}$$

and equality occurs if and only if M is isometric to a standard sphere and $r = \text{Inj}(M)$.

The bound $a(d)$ in theorem 149 on the preceding page is certainly not optimal, in that equality can never be attained. It is not clear if it is reasonable to conjecture for the optimal bound the value obtained for standard hemisphere, namely

$$a(d) = \frac{\sigma(d)}{(\pi/2)^d}$$

with equality only for standard hemispheres.

Definition 151 For a compact manifold M of dimension d its embolic constant¹⁴ is

$$\text{Emb}(M) = \inf_g \frac{\text{Vol}(M, g)}{\text{Inj}(M, g)^{\dim M}}$$

where the infimum is taken over all Riemannian metrics g on M .

By theorem 148 on the previous page we know that this is a positive constant which is not smaller than

$$\frac{\sigma(d)}{\pi^d}.$$

Imagine the subset of $\mathbb{R}^+$ made up by the different embolic constants of the various compact manifolds of a given dimension d . Notice the major result: theorem 267 on page 505 which is a direct corollary of theorem 377 on page 622.

Question 152 Compute $\text{Emb}(M)$ for various standard compact manifolds e.g. the tori T^d , the various $\mathbb{K}\mathbb{P}^n$, etc. In case one is able to compute the embolic constant, we may also wonder for which metric is it attained?

Question 153 Is the value

$$\frac{\sigma(d)}{\pi^d}$$

isolated or not in the set of possible embolic constants for manifolds of a given dimension?

¹⁴ The term *embolism* comes from the Greek for *insertion* and refers in physiology to a bubble or blood clot blocking an artery. We want to think of the injective image of a ball under the exponential map as our embolism, “blocking” the manifold in the sense that it strikes into itself, and this is the relevant analogy.

Question 152 is completely open today, even for projective spaces $\mathbb{RP}^d$ or tori. Except for $\text{Emb}(S^d)$ not a single embolic constant $\text{Emb}(M)$ is known when $d \geq 3$. As an exercise, the reader can find the embolic constants of $\mathbb{RP}^2$, the two dimensional torus and the Klein bottle (use Klingenberg's theorem 89 on page 272). If the embolic constant of $\mathbb{K}\mathbb{P}^n$ turns out to be the one obtained for the standard metric then this will help to answer (and probably completely answer) the outrageously open question 97 on page 285 concerning Riemannian manifolds M with

$$\text{diam}(M) = \text{Inj}(M).$$

But there is a satisfactory answer to question 153:

Theorem 154 (Croke 1988 [418]) *Every manifold M of dimension d which is not a sphere satisfies*

$$\text{Emb}(M) > \text{Emb}(S^d) + c(d)$$

where $c(d)$ is an explicitly described positive constant depending only on the dimension d .

Note 7.2.4.1 (Best metrics) We have introduced a remarkable invariant of (smooth) compact manifolds: the embolic constant. This invariant comes from Riemannian geometry. In chapter 11 we will introduce two other ones:

- the minimal volume and
- the infimum of the integral

$$\int_M \|R\|^{d/2} dV_M.$$

Riemannian metrics achieving the lowest bound for a reasonable functional are, in some sense, the best possible on a given compact manifold. $\blacklozenge$

7.2.4.2 Starting the Proof and Introducing the Unit Tangent Bundle

We will give the ideas for the proofs of theorems 148 on page 355, 149 on page 355, 150 on page 355, and 154 when the manifold in question is a surface. We will just mention the technical difficulties for passing from surfaces to higher dimensional manifolds. But the main steps are already present in studying surfaces. In passing, we will now introduce the unit tangent bundle of a Riemannian manifold, which will also play various important roles in the future (for example, see chapter 10).

First, normalize the metric g in order that

$$\text{Inj}(M, g) = \pi.$$

We consider various balls $B(m, r)$ of radii $r < \pi$. We can be sure that these balls are nicely covered by the exponential map. In such a ball, the ds^2 is written as

$$ds^2 = dt^2 + f^2(u, t) du^2.$$

Consequently, we see

$$\text{Area } B(m, r) = \int_{u \in U_m M} \int_0^r f(u, t) dt dV_{U_m M}(u) \quad (7.5)$$

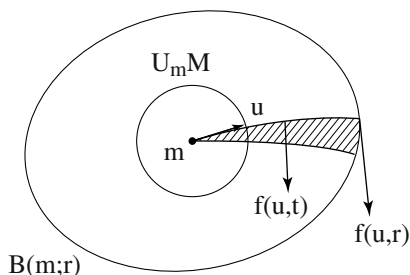


Fig. 7.42. Measuring the volumes of balls via the exponential map

Moreover, we can consider u as a unit tangent vector at m running through the unit sphere $U_m M$. If we denote by γ_u the geodesic with

$$\gamma'_u(0) = u$$

then the function f is given along γ_u by the Gauß–Jacobi equation

$$f'' + Kf = 0. \quad (7.6)$$

The general idea is to take some kind of “mean value” of areas of all the balls when m runs through M and r ranges from 0 to π . This will involve four integrations:

1. in t from 0 to r
2. in r from 0 to π
3. in u running through $U_m M$ and finally
4. in m running through M .

The last two integrations are better packaged into one by introducing

Definition 155 *The set of all unit tangent vectors to a Riemannian manifold M is called its unit tangent bundle and is denoted by UM . It is a subbundle of the full tangent bundle TM . The canonical projection is denoted by*

$$p : UM \rightarrow M.$$

The bundle UM is automatically oriented, and endowed with a canonical Riemannian metric (see §15.2) which moreover makes $p : UM \rightarrow M$ a Riemannian submersion (see §§4.3.6). In particular, it has a canonical volume form which is denoted by ω . The geodesic flow on UM is the one parameter group G_t of diffeomorphisms of UM defined as follows:

$$G_t(u) = \gamma'_u(t).$$

Theorem 156 (Liouville) *The volume form ω is invariant under the geodesic flow (see chapter 10).*

In its canonical metric, the trajectories of the geodesic flow are themselves geodesics and (see on page 237) project down onto the geodesics of M .

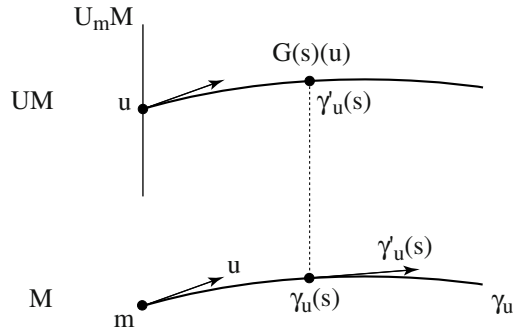


Fig. 7.43. The geodesic flow lines in UM project to the geodesics of M

A physicist might consider UM as the phase space of free particles on M . It is helpful to look at the geodesics as trajectories of a flow in UM because in UM they fill up the space and do not intersect each other. This will turn to be very important in chapter 10.

7.2.4.3 The Core of the Proof

The next idea in proving the theorems 148 on page 355, 149 on page 355, 150 on page 355, and 154 on page 357 is to apply Fubini's theorem many times. The starting point is to integrate things (i.e. the various functions figuring in equation 7.5 on the preceding page) first along a given geodesic segment of length π . To do this, we decorate the function f with a subscript as f_u when we want to describe its restriction to γ_u . We have to compute the double integral :

$$I(u) = \int_{s=0}^{s=\pi} ds \int_{t=s}^{t=\pi} dt f(G_s(u), t) \quad (7.7)$$

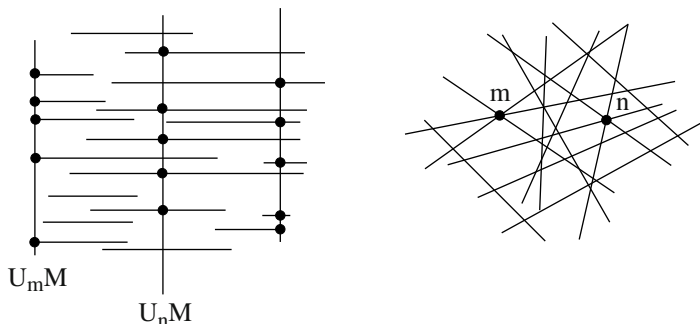


Fig. 7.44. (a) Geodesics in UM upstairs (b) Geodesics downstairs in M

The basic idea of the proof is that the various $f(G_s(u), t)$ (when s varies) can be explicitly computed as a function of $f(u, t)$. This is because they all are solutions of the Gauß equation

$$f'' + Kf = 0 \quad (7.8)$$

It is essential here that $f(u, t)$ does not vanish between 0 and π . This is because of the results on cut points which we found in §§6.5.1. Now elementary theory of ordinary differential equations yields

$$f(G_s(u), t) = f(s)f(t) \int_{w=s}^{w=t} \frac{dw}{f^2(w)} \quad (7.9)$$

where

$$f(s) = f(u, s).$$

We might guess that there will some kind of compensation of the following sort. Assume for example that f is very small on a large interval. If f were moreover a constant then this will imply $K = 0$. But then, on such an interval, $f(G_s(u), t)$ will be linear and as such will get large. This guess is correct

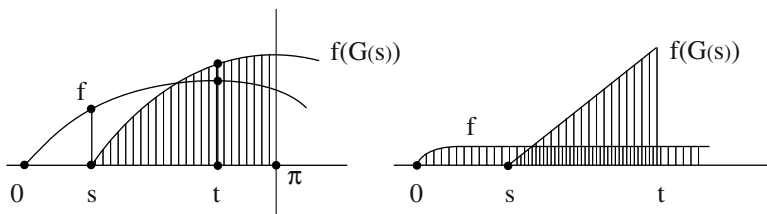


Fig. 7.45. Behaviour of the Jacobi field

and the sphere yields the minimal compensation since

Lemma 157 (Basic lemma)

$$I(u) \geq \pi$$

with equality only if f is proportional to $\sin t$, i.e. if

$$K \equiv 1.$$

The proof is very clever, but completely elementary; see Berger 1977 [156] if you are stuck. What you have to prove is that for any function ϕ vanishing at 0 and π

$$\int_{x=0}^{x=\pi} dx \int_{y=x}^y y = s dy \int_{t=x}^{t=y} dt \frac{\phi(x)\phi(y)}{\phi^2(t)} \geq \frac{s^3}{\pi^2}$$

and equality happens if and only if ϕ is proportional to the sine function.

The proof of theorem 148 on page 355 is now finished as follows. To every $u \in UM$ one attaches the picture in figure 7.46.

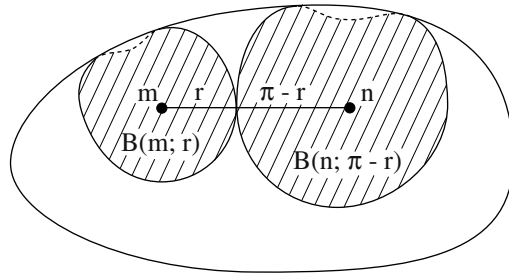


Fig. 7.46. Two balls sitting astride each other

By the definition of the injectivity radius, for every $u \in UM$ and every $r < \pi$, the two balls

$$B(m = p(u), r) \quad \text{and} \quad B(n = p(G_\pi(u)), \pi - r)$$

are disjoint. This show that

$$\text{Area}(M) \geq \text{Area } B(m, r) + \text{Area } B(n, \pi - r). \quad (7.10)$$

We integrate this inequality on the whole of UM , and for r running from 0 to $\pi/2$, using equation 7.5 on page 358. Then we apply the Fubini theorem and on the way apply Liouville's theorem 156 on page 359. and finally lemma 157. This yields theorem 148 on page 355. It is easy to identify the manifold when we have equality in our inequality: we are forced to have equality in lemma 157 and this forces the curvature to be constant and equal to 1. We are then on a sphere: apply the results of §6.3.2 and we remark moreover that M cannot be a nontrivial quotient of the standard sphere since its injectivity radius is

$$\text{Inj}(M) = \pi.$$

When the dimension is higher than two, as we have seen in equation 6.11 on page 248 the Gauß ordinary differential equation

$$f'' + Kf = 0$$

has to be replaced by the vector Jacobi equation

$$Y'' + R(\gamma', Y)\gamma' = 0$$

and the function f by the determinant

$$\det(Y_1, \dots, Y_{d-1})$$

of Jacobi fields. We replace the scalar equation 7.9 on page 360 above by the following one, which an endomorphism equation (or a matrix equation, if you prefer)¹⁵

$$A(G(s)(u), t) = A(s) \left(\int_s^t (A^t(r) A(r))^{-1} dr \right) A^t(s)$$

For this relation, as well as for a detailed proof, one can consult for example section VI.2 of Sakai's book Sakai 1996 [1085], or appendix D of Besse 1978 [182]. With a clever trick, the classical inequality for integrals of convex functions, one has finally to replace the basic lemma 157 on the previous page by the following inequality:

Lemma 158 (Basic inequality) *If*

$$\rho : [0, \pi] \rightarrow \mathbb{R}^+$$

is any function satisfying

$$\rho(\pi - t) = \rho(t)$$

then for any nonnegative function ϕ

$$\int_0^\pi dr \int_r^\pi ds \int_r^t \frac{\phi(r)\phi(t)}{\phi^2(s)\rho(t-r)} \geq \text{"the value the same expression takes for } \phi = \sin \text{"}$$

and with equality if and only if ϕ is proportional to the sine function.

This very subtle inequality was proven by Kazdan in 1978: see section 5.3 of Chavel 1993 [326] or appendix E of Besse 1978 [182] or section VI.2 of Sakai 1996 [1085]. It is an open problem to find a proof of it with a clear underlying concept; one would expect interesting generalizations and applications of such a proof.

¹⁵ Clearly, the order of matrix products is significant.

7.2.4.4 Croke's Three Results

We now sketch the proof of theorem 149 on page 355. The proof is a mixture of the basic inequality in lemma 158 on the preceding page with Santalo's equation 7.11 on the following page which was already used in §§§7.1.2.3 by Croke to prove the conjecture for the isoperimetric profile of four dimensional manifolds of nonpositive curvature. Santalo's formula is proven in section 5.2 of Chavel 1993 [326].

The proof is concerned with balls of radius $r < \pi/2$. The sphere which is the boundary of this ball will be denoted

$$S(m, r) = \partial B(m, r).$$

This sphere is smooth because we are within the injectivity radius. Moreover, for any point $n \in B(m, r)$, the injectivity radius at n , $\text{Inj}(n)$, is not smaller than π , which has the effect that the geodesics through n will nicely sweep out over the whole ball $B(m, r)$. And there will be no conjugate points on these geodesics inside $B(m, r)$.

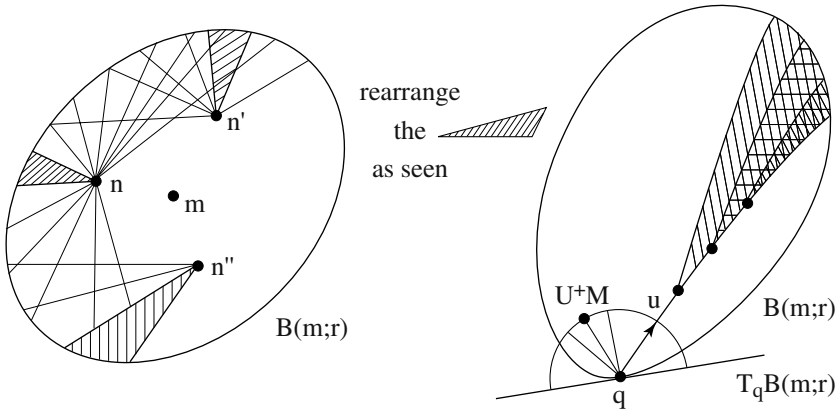


Fig. 7.47. (a) Sweeping out the whole ball about m from any point n inside it. (b) Sweeping out geodesics inside the ball from a point of the boundary.

This implies that we can compute

$$\text{Vol } B(m, r)$$

from the viewpoint of n , using an integral over $U_n M$ of suitable determinants of Jacobi fields. We now carry out these integrals over all n running through $B(m, r)$ and exchange the order of integration to integrate along the various geodesics γ_u where u runs through the unit tangent vectors $U_q^+ M$ where, for a point of the sphere $q \in S(m, r)$, the half tangent unit sphere $U_q^+ M$ is that made of the vectors pointing inside $B(m, r)$.

We use now Kazdan's inequality from lemma 158 on page 362 and Liouville's theorem just as in Santalo's formula below. Consider in some compact Riemannian manifold M a precompact piece of smooth hypersurface N having a side, so that we can talk about U^+N . Consider an arbitrary positive function $\psi(u)$ on U^+N , but small enough that

$$\psi(u) < \text{Cut}(u)$$

(smaller than the cut value) and any integrable function f defined on the set D made up by the geodesic flows $G(s)(u)$ where u runs through U^+N , and s runs through the interval $[0, \psi(u)]$. Denote by $\cos(u)$ the cosine of the angle between the vector u and the normal vector to N . Then one has Santalo's formula

$$\int_D f(v) dv = \int_{U^+N} \left(\int_0^{\psi(u)} f(G_s(u)) \cos(u) ds \right) dV_{U^+N}(u) \quad (7.11)$$

The proof is nothing but Fubini's theorem and the change of variables formula; for details see page 286 of Berger 1965 [151] or Santalo 1976 [1093].

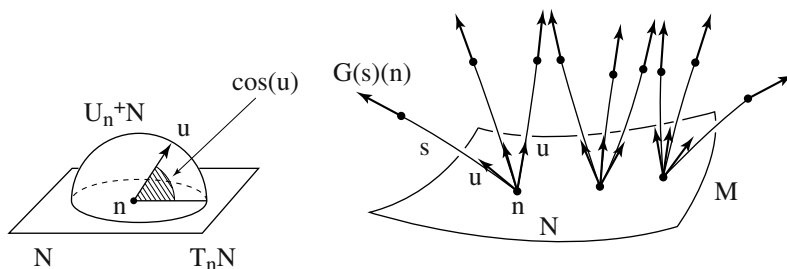


Fig. 7.48. Santalo's formula in pictures

All together this results in the following inequality between the volume of the ball $B(m, r)$ and that of its boundary $S(m, r)$:

$$\text{Vol } S(m, r)^d \geq a(d) \text{Vol } B(m, r)^{d-1} \quad (7.12)$$

But

$$\text{Vol } S(m, r) = \frac{\partial}{\partial r} \text{Vol } B(m, r)$$

so that this last inequality is a differential inequality which by integration from 0 to $\pi/2$ furnishes theorem 149 on page 355. That this inequality is certainly not optimal is left to the reader. The proof of theorem 150 on page 355 is a refinement of that of theorem 149, still largely using only Santalo's formula and the basic inequality.

The proof of theorem 154 on page 357 now follows theorem 149 inequality with the following elegant modification:

Lemma 159 *Assume that M is not a sphere. Then for every $u \in UM$ and every $r \in [0, \pi/2]$ there is some $q \in M$ such that*

$$B(\gamma_u(0), r) \cap B(\gamma_u(\pi), \pi - r) \cap B(q, r) = \emptyset$$

and in particular

$$\text{Vol } M \geq \text{Vol } B(\gamma_u(0), r) + \text{Vol } B(\gamma_u(\pi), \pi - r) + \text{Vol } B(q, r).$$

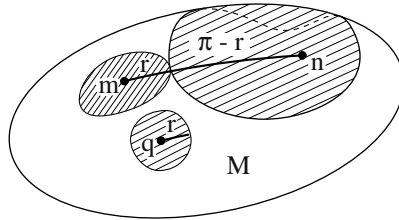


Fig. 7.49. Three balls that don't fill up all the whole manifold

Croke's result is obtained by contradiction, resting on the folk theorem that if a manifold can be covered by only two balls, then it is a sphere: see for example theorem 4.1 of Milnor 1963 [921]; this theorem is essential in the proof of theorem 297 on page 552. But assume there is no such point q . This means (by the triangle inequality) that M is covered by the two balls

$$B(\gamma_u(0), 2r) \text{ and } B(\gamma_u(\pi), \pi).$$

When one integrates the inequality in lemma 159 over all unit vectors u , as in the proof in §§7.2.4.3, the two first terms yield exactly the volume of M but the third terms—thanks to theorem 149 on page 355—add up to a positive constant which is universal in the dimension d and this is exactly theorem 154 on page 357.

Such an inequality is certainly not optimal since the proof leaves a lot of room between the third ball and the other ones. Contrarily, for the standard spheres the two first balls never leave any room for a third.

Note 7.2.4.2 As Croke remarks in his work, the above proof admits with no modification the following generalization. The *category* of a compact manifold is the minimum number of topological balls needed to cover it. By the way, the category of a d dimensional manifold never exceeds $d + 1$. The desired result is

$$\text{Emb}(M) \geq \text{Emb}(S^d) + c(d)(\text{category}(M) - 2).$$

See more on this in §§11.2.3.



7.2.4.5 Infinite Injectivity Radius

From corollary 55 on page 228 and definition 85 on page 268 we know that the exponential map

$$\exp_m : T_m M \rightarrow M$$

is a covering for every $m \in M$ if and only if M has no conjugate points (for any interval and on any geodesic). For example, this happens on a manifold of nonpositive curvature; see §§ 6.3.3 on page 254. But there are Riemannian manifolds with no conjugate points but with parts having positive curvature (not too much).

For various reasons (some of them are explained in chapter 12), Riemannian manifolds without conjugate points are of great interest. Let M be such a manifold and consider its universal Riemannian covering $\tilde{M}$. It is simply $T_m M$ endowed with the pullback metric $\exp_m^* g$: see the proof of corollary 55 on page 228. Now of course $\tilde{M}$ is not compact, but is complete and its injectivity radius is infinite. In particular, for every positive r and every point $\tilde{m}$ the ball $B(\tilde{m}, r)$ is diffeomorphic to $\mathbb{R}^d$ and nicely covered by the geodesics emanating from $\tilde{m}$. Since the deck transformations are isometries, in particular two points $\tilde{m}, \tilde{n} \in \tilde{M}$ having the same projection $m \in M$ yield balls $B(\tilde{m}, r)$ and $B(\tilde{n}, r)$ which have the same volume. So one can refer without ambiguity to the volume

$$\text{Vol } B(m, \tilde{M}, r)$$

as if it were attached to the point $m \in M$ downstairs.

When the curvature is nonpositive, we have seen in theorem 103 on page 306 that the volume of balls at any point grow with their radius r as fast as in Euclidean space $\mathbb{E}^d$. Can one expect the same conclusion, but with the weaker hypothesis that the manifold has no conjugate points? In 1991, Croke 1992 [421] discovered that this is the case. There are two results: a pointwise one and an averaging one:

Theorem 160 (Croke 1991 [421]) *Let M be a compact manifold without conjugate points. Then for every $r > 0$*

$$\frac{1}{\text{Vol } M} \int \text{Vol } B(m, \tilde{M}, r) \, dr \geq \omega_d r^d$$

with equality if and only if M is flat (where ω_d is the volume of the unit ball in d dimensional Euclidean space).

Theorem 161 *For every $r > 0$ and for every point $m \in M$*

$$\liminf_{r \rightarrow \infty} \frac{\text{Vol } B(m, \tilde{M}, r)}{\omega_d r^d} \geq 1$$

with equality if and only if M is flat.

These results are applied in §10.11. The proofs of the above results use some of the same inequalities we used above but also much more sophisticated ones and subtle limit arguments. We refer the reader to Croke's paper.

7.2.4.6 Using Embolic Inequalities and Local Contractibility

Besides being very natural in the study of relations between curvature and topology, results of embolic nature are essential for some other reasons. We mention them here briefly before meeting them again later on.

Embolic results permit good, even optimal, constants in Sobolev inequalities; see theorem 118 on page 325. and the survey Hebey 1996 [690]. Various inequalities for the eigenvalues of the Laplace operator make essential use of Croke's local inequality from theorem 149 on page 355. One also finds embolic inequalities in the C^∞ compactness of isospectral moduli (see §§9.12.3). An optimal embolic inequality for $\mathbb{CP}^n$ will solve the question of §§10.4.2. The optimal inequality in theorem 148 on page 355 is used to solve the Wiedersheimannigfaltigkeiten problem; see theorem 257 on page 493.

The essential property of (open) balls of radius smaller than or equal to the injectivity radius is that they are contractible. For example the inequality in theorem 149 on page 355 has the following immediate corollary (use the covering trick of lemma 125 on page 333):

Theorem 162 *A compact manifold of a given dimension d can be covered by a number of contractible sets, and this number is universally bounded in terms of the total volume and the manifold and its injectivity radius.*

We will see in §§12.4.1.2 how important this corollary is. But luckily one can do surprisingly many things with a weaker notion of contractibility, which first appeared in Gromov 1981 [613]:

Definition 163 *A contractibility function for a metric space is a function*

$$\rho : [0, R] \rightarrow \mathbb{R}^+$$

with $R > 0$, so that

$$\rho(r) \rightarrow 0 \text{ as } r \rightarrow 0$$

and

$$\rho(r) \geq r$$

for all $r \in [0, R]$. A metric space is said to be locally geometrically contractible with contractibility function ρ , written symbolically $\text{LGC}(\rho)$, if every metric ball $B(x, r)$ is contractible in the ball $B(x, \rho(r))$ for every point x and every radius $r \leq R$. The space is said to be geometrically contractible if the function ρ is actually defined on all $\mathbb{R}^+$ and the space is $\text{LGC}(\rho)$ for ρ restricted to $[0, R]$ for any number $R > 0$.

Thinking along the lines of theorems 148 on page 355 and 149 on page 355, we would like to have analogies of these results in the realm of local or global contractible metric spaces. For (complete) noncompact Riemannian manifolds, on page 43 of Gromov 1983 [618] it is proven that geometric contractibility implies, via infinite filling volume, infinite volume. The local version of this sort of result for $\text{LGC}(\rho)$ Riemannian manifolds was proven in Greene & Petersen 1992 [592]: in a complete noncompact $\text{LGC}(\rho)$ manifold¹⁶, the volume of balls grows at least linearly; but if moreover

$$\rho(r) \geq cr$$

then the volume of balls of radius r grows at least as fast as

$$\text{Vol } B(m, r) \geq Cr^d.$$

The concept of *filling radius* (defined on page 342) plays an important role. We will find the $\text{LGC}(\rho)$ condition entering essentially in §§§12.4.1.2. This notion is becoming increasingly important in various spaces more general than Riemannian manifolds; for example see Semmes 1996 [1122].

¹⁶ With R finite; otherwise it wouldn't be interesting.